

DigSig2

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This summary follows the lectures of Prof. Dr. G. Schuster and the book "Introduction to Signal Processing" by Sophocles J. Orfanidis. Most figures and formulas are taken from there.

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1. Digital Filter Realizations Chapter 7

1.1. Direct Form S265

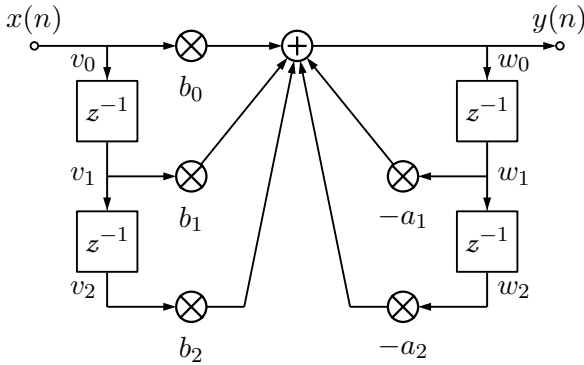
Also called *direct form I*. Consider a second-order filter with transfer function

$$H(z) = \frac{N(z)}{D(z)} = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{1 + a_1 z^{-1} + a_2 z^{-2}}$$

with the I/O difference equation

$$y_n = -a_1 y_{n-1} - a_2 y_{n-2} + b_0 x_n + b_1 x_{n-1} + b_2 x_{n-2}$$

This results in the following block diagram:



and following sample processing algorithm:

```

for each input sample x do :
    v0 = x
    w0 = -a1w1 - a2w2 + b0v0 + b1v1 + b2v2
    y = w0
    v2 = v1, w2 = w1
    v1 = v0, w1 = w0
  
```

Filter Order The order of a digital filter is equal to the highest power of z^{-1} appearing in the denominator polynomial.

$$H(z) = \frac{N(z)}{D(z)} = \frac{b_0 + b_1 z^{-1} + \dots + b_L z^{-L}}{1 + a_1 z^{-1} + \dots + a_M z^{-M}}$$

The filter order is therefore

Order = M.

The numerator may have a lower order than the denominator.

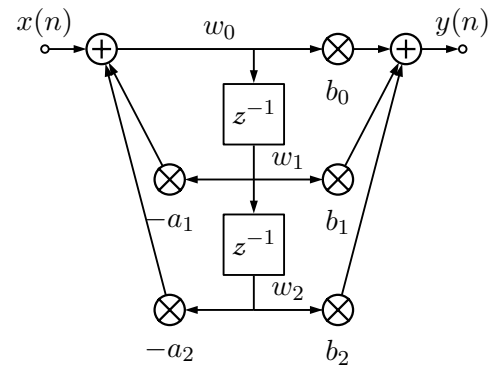
1.2. Canonical Form S271

The most popular form. Also called *direct form II*. It can be obtained from the Direct Form S265 by splitting it up, introducing an intermediate signal $w(n)$ and reversing it.

$$H(z) = \frac{1}{D(z)} \cdot N(z)$$

$$w(n) = x(n) - a_1 w(n-1) - a_2 w(n-2)$$

$$y(n) = b_0 w(n) + b_1 w(n-1) + b_2 w(n-2)$$



for each input sample x do :

```

w0 = x - a1w1 - a2w2
y = b0w0 + b1w1 + b2w2
w2 = w1
w1 = w0
  
```

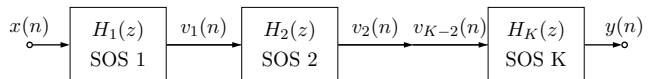
1.3. Cascade Form S277

For filters with real impulse responses (all a_i and b_i are real), the transfer function can be written by cascaded 2nd order sections (SOS).

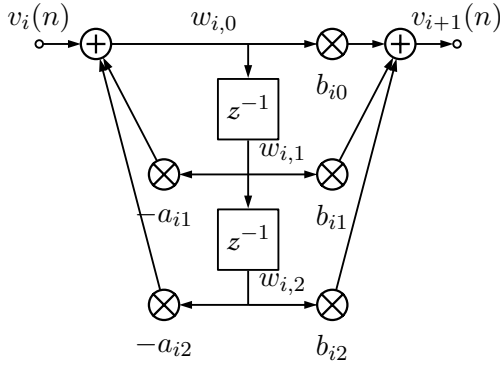
$$H(z) = \prod_{i=0}^{K-1} H_i(z) = \prod_{i=0}^{K-1} \frac{b_{i0} + b_{i1} z^{-1} + b_{i2} z^{-2}}{1 + a_{i1} z^{-1} + a_{i2} z^{-2}}$$

Usually the SOS are implemented in the canonical form. A first order term is implemented by setting the z^{-2} term to zero. The advantage of the cascade form is the efficient implementation of 2nd order sections in DSPs.

Block Diagram The K second-order sections are connected in series:



Each second-order section is implemented in the canonical form (Direct Form II):



For a first-order section, set

$$a_{i2} = 0, \quad b_{i2} = 0.$$

$$h(n) = b(n) * \sum_{k=0}^{\infty} \delta(n - kD),$$

which means that the FIR impulse response $b(n)$ is repeated every D samples.

Example:

$$H(z) = \frac{1 + 2z^{-1}}{1 - z^{-4}}$$

results in

$$h = [1, 2, 0, 0, 1, 2, 0, 0, 1, 2, 0, 0, \dots].$$

1.4. Cascade $\Leftrightarrow$ Canonical S284

To find the Cascade Form S277 of a filter, $D(z)$ and $N(z)$ are factored into 2nd order sections.

$$\begin{aligned} D(z) &= 1 + a_1 z^{-1} + a_2 z^{-2} + \dots + a_M z^{-M} \\ &= (1 - p_1 z^{-1})(1 - p_2 z^{-1}) \dots (1 - p_M z^{-1}) \end{aligned}$$

For real-valued roots, any two factors can be combined:

$$(1 - p_1 z^{-1})(1 - p_2 z^{-1}) = (1 - (p_1 + p_2)z^{-1} + p_1 p_2 z^{-2})$$

Conjugate complex roots are paired (if the coefficients are real):

$$(1 - p_1 z^{-1})(1 - p_1^* z^{-1}) = 1 - 2\text{Re}(p_1)z^{-1} + |p_1|^2 z^{-2}$$

1.5. Impulse Response of Periodic Filters

Many transfer functions in this chapter contain a denominator of the form

$$1 - z^{-D}.$$

Using the geometric series,

$$\frac{1}{1 - z^{-D}} = \sum_{k=0}^{\infty} z^{-kD} = 1 + z^{-D} + z^{-2D} + \dots$$

the impulse response can be obtained without using partial fraction expansion.

If

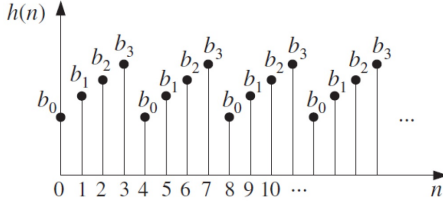
$$H(z) = \frac{B(z)}{1 - z^{-D}},$$

then

2. Signal Processing Applications Chapter 8

2.1. Digital Waveform Generators S316

The goal is to design a filter $H(z)$ with a desired impulse response $h(n)$.



This waveform can be sample-by-sample computed as the impulse response of the filter $H(z)$ or it can be pre-computed and stored in memory (wavetable). If the waveform is periodic (often) then the fundamental frequency can be controlled by the speed the wavetable is read out.

2.1.1. Sinusodial Generators S316-320

A causal sinusoidal is implemented by

$$h(n) = R^n \sin(\omega_0 n) u(n)$$

$$H(z) = \frac{R \sin \omega_0 z^{-1}}{1 - 2R \cos \omega_0 z^{-1} + R^2 z^{-2}}$$

with $\omega_0 = 2\pi f_0/f_s$ being the digital frequency.

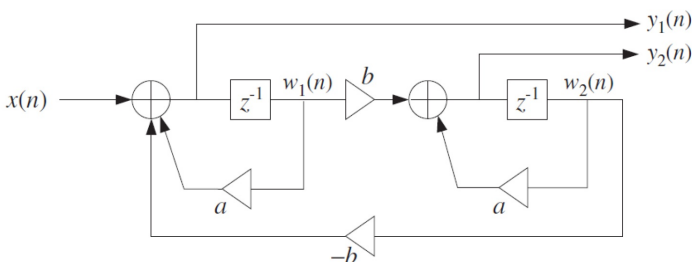
For $R = 1$, $h(n)$ is a pure sine, for $0 < R < 1$, $h(n)$ is a geometrically declining sinusoidal wave.

A cosinusoidal signal is generated similarly with

$$h(n) = R^n \cos(\omega_0 n) u(n)$$

$$H(z) = \frac{1 - R \cos \omega_0 z^{-1}}{1 - 2R \cos \omega_0 z^{-1} + R^2 z^{-2}}$$

As the poles of the transfer functions are at conjugate complex locations, both generators can be combined to the *coupled form*:



with

$$a = R \cos \omega_0 \quad b = R \sin \omega_0$$

or in the z -domain:

$$H_1(z) = \frac{1 - a z^{-1}}{1 - 2a z^{-1} + (a^2 + b^2) z^{-2}} \quad (\text{cosine})$$

$$H_2(z) = \frac{b z^{-1}}{1 - 2a z^{-1} + (a^2 + b^2) z^{-2}} \quad (\text{sine})$$

2.1.2. Periodic Waveform Generators S321-329

A sampled version of a periodic signal is not necessarily periodic. In order for $x(n)$ to be periodic in n with a period of D samples, one whole period of the signal must fit within the D samples. Therefore at $n = D$ the signal must cycle by one whole period.

This requires that $x(D) = x(0)$, which requires

$$\omega = \frac{2\pi}{D} \quad f = \frac{f_s}{D} \quad f_s = Df \quad T_D = DT$$

Because of the periodicity, it is enough to specify the signal over one period only:

$$h = [b_0, b_1, \dots, b_{D-1}, b_0, b_1, \dots, b_{D-1}, b_0, \dots]$$

With the corresponding z -transform:

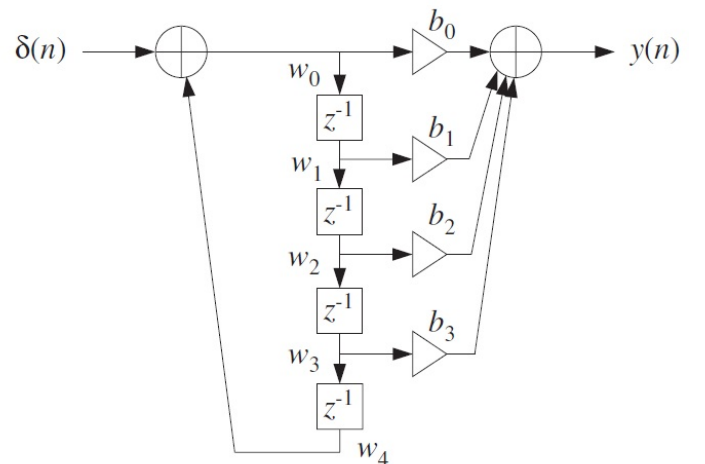
$$H(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots + b_{D-1} z^{-(D-1)}}{1 - z^{-D}}$$

or the impulse response

$$h(0) = b_0 \quad h(1) = b_1 \quad \dots \quad h(D-1) = b_{D-1}$$

$$h(n) = h(n-D) \quad \text{for } n \geq D$$

The canonical realization is the following:



Example (Initialization) For

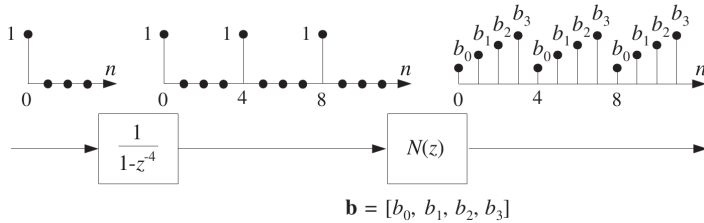
$$H(z) = \frac{z^{-1} + 2z^{-2}}{1 - z^{-3}}$$

the canonical realization is initialized by exciting the filter with a unit impulse $x(0) = 1$.

n	w_0	w_1	w_2	w_3	y
0	1	0	0	0	0
1	0	1	0	0	1
2	0	0	1	0	2
3	1	0	0	1	0
4	1	1	0	0	1
5	0	1	1	0	2

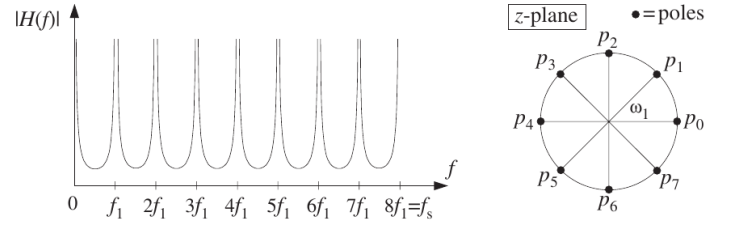
The sequence stored in the recursive delay line repeats every $D = 3$ samples and therefore generates the periodic waveform.

The common delay line of the canonical form can be split up into two identical delay lines. The first recursive filter creates a periodic impulse train as impulse response, which in turn creates a periodic waveform which consists of the finite impulse response of the FIR filter excited over and over again



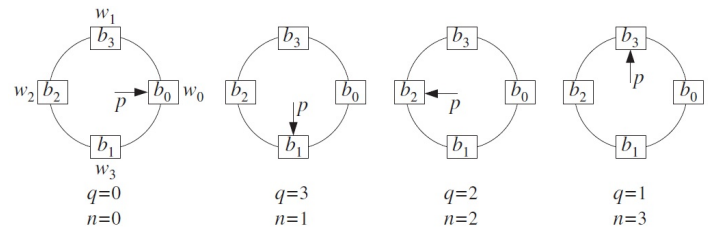
These sequences are causal, therefore they have a comb-like spectra, with dominant peaks (poles) at the harmonics. The spectrum can be obtained by setting $z = e^{j\omega}$ in the transfer function:

$$\begin{aligned}
 1 - z^{-D} &= \dots = 2je^{j\omega D/2} \sin\left(\frac{\omega D}{2}\right) \\
 |H(\omega)| &= \frac{|N(\omega)|}{|1 - e^{-j\omega D}|} = \frac{|N(\omega)|}{2|\sin(\frac{\omega D}{2})|} \\
 \omega &= 2\pi f / f_s \\
 |H(f)| &= \frac{|N(f)|}{2 \underbrace{|\sin(\frac{\pi f D}{f_s})|}_{=0}} \\
 &\Rightarrow \frac{\pi f D}{f_s} = \pi m \\
 \Leftrightarrow f_m &= m \frac{f_s}{D} = m f_1 \quad f_1: \text{fundamental frequency}
 \end{aligned}$$



2.1.3. Wavetable Generators S330-349

An alternative approach is to store the samples of a period in memory and play this sample over and over again. This is called a wavetable $\mathbf{w}$. The readout is accomplished using a circular pointer p and, if needed, an offset index q .



The wave can easily be delayed by m time units by setting

$$y(n) = b(n - m)$$

This is achieved by initializing p and q with

$$p = w + m \quad q = m$$

This leads to the following sequences for different m 's

$$\begin{aligned}
 m = 0 & \quad [b_0, b_1, b_2, b_3, b_0, b_1, b_2, b_3, b_0, b_1, b_2, b_3, \dots] \\
 m = 1 & \quad [b_3, b_0, b_1, b_2, b_3, b_0, b_1, b_2, b_3, b_0, b_1, b_2, \dots] \\
 m = 2 & \quad [b_2, b_3, b_0, b_1, b_2, b_3, b_0, b_1, b_2, b_3, b_0, b_1, \dots] \\
 m = 3 & \quad [b_1, b_2, b_3, b_0, b_1, b_2, b_3, b_0, b_1, b_2, b_3, b_0, \dots]
 \end{aligned}$$

The wavetable information can be manipulated in many possible ways to achieve particular effects. A wavetable of length D has a fundamental frequency f of $f = \frac{f_s}{D}$. A shorter period $d \leq D$, meaning a higher frequency $f = \frac{f_s}{d}$ can easily be achieved by skipping samples.

The constant $c = \frac{D}{d}$ describes, how often the sub-period d fits into the original period D . To keep f within the symmetric Nyquist interval, c has to be inside the range

$$-\frac{D}{2} \leq c \leq \frac{D}{2}$$

The update of the offset q is

$$q_{n+1} = (q_n - c) \% D$$

Non-integer offsets can be truncated, rounded or, if higher quality is needed, interpolated.

2.1.4. Fractional Wavetable Readout

If the read pointer is not an integer, the neighbouring samples must be used to determine the output value.

Let

$$p = i + \alpha, \quad 0 \leq \alpha < 1,$$

where $i = \lfloor p \rfloor$ is the integer part and α the fractional part.

The most common readout methods are:

Truncate Down Always use the lower index.

$$y = w_i$$

Truncate Up Always use the upper index.

$$y = w_{i+1}$$

Round Use the nearest sample.

$$y = \begin{cases} w_i, & \alpha < 0.5 \\ w_{i+1}, & \alpha \geq 0.5 \end{cases}$$

Linear Interpolation Use a weighted average of the neighbouring samples.

$$y = (1 - \alpha)w_i + \alpha w_{i+1}$$

Linear interpolation provides the smoothest waveform and therefore introduces the smallest distortion.

Example For

$$w = [3, 5, 8, 6, 4], \quad p = 1.25$$

Method	Formula	Output
Truncate Down	$y = w_i$	5
Truncate Up	$y = w_{i+1}$	8
Round	$\alpha < 0.5$	5
Linear Interpolation	$y = (1 - \alpha)w_i + \alpha w_{i+1}$	5.75

2.2. Noise Reduction and Signal Enhancement S382

Goal: extracting a signal from a noisy measurement.

$$\text{Noisy measurement} \quad x(n) = s(n) + \nu(n)$$

$$\text{Desired signal} \quad s(n)$$

$$\text{Undesired noise} \quad \nu(n)$$

2.2.1. Noise Reduction Filters S382-386

The ideal noise reduction filter creates the output

$$y(n) = y_s(n) + y_\nu(n) \quad \text{with} \quad \begin{aligned} y_s(n) &= s(n) \\ y_\nu(n) &= 0 \end{aligned}$$

The power spectral density of the output is calculated as

$$S_{yy}(\omega) = |H(\omega)|^2 S_{xx}(\omega)$$

with

$$|H(\omega)|^2 = H(\omega) \cdot H^*(\omega)$$

and with white noise as the input $S_{xx} = \sigma_x^2$

$$S_{yy}(\omega) = |H(\omega)|^2 \sigma_x^2$$

Note that the output noise is not white anymore and is therefore colored noise.

The *noise reduction ratio* NRR measures the noise power at the output compared to the noise power at the input. Note that a small NRR means large noise reduction.

$$\text{NRR} = \frac{\sigma_y^2}{\sigma_x^2} = \int_{-\pi}^{\pi} |H(\omega)|^2 \frac{d\omega}{2\pi} = \sum_n h(n)^2$$

The *signal to noise ratio* SNR can be calculated at the input and at the output, hence the relative improvement of the SNR is

$$\frac{\text{SNR}_{\text{out}}}{\text{SNR}_{\text{in}}} = \frac{1}{\text{NRR}} \cdot \frac{E[y_s(n)^2]}{E[s(n)^2]}$$

or if the desired signal $s(n)$ is not changed by the filter

$$\frac{\text{SNR}_{\text{out}}}{\text{SNR}_{\text{in}}} = \frac{1}{\text{NRR}}$$

The best possible NRR is achieved by using ideal filters:

$$\text{Ideal lowpass filter} \quad \text{NRR} = \frac{\omega_c}{\pi}$$

$$\text{Ideal bandpass filter} \quad \text{NRR} = \frac{\omega_b - \omega_a}{\pi}$$

Most real noise reduction filters are linear phase filters, where each frequency is delayed by the same number of samples and implies constant phase delay:

$$y_s(n) = s(n - D)$$

First-order IIR Smoother S386 To remove zero-mean white Gaussian noise of variance σ_v^2 , an IIR lowpass filter is used:

$$H(z) = \frac{b}{1 - a z^{-1}}$$

With $0 < a < 1$ and $H(1) = 1$, b can be fixed to $b = 1 - a$. In that case the NRR is

$$\text{NRR} = \frac{b^2}{1 - a^2} = \frac{1 - a}{1 + a}$$

The transient time constant will be

$$n_{\text{eff}} = \frac{\ln \epsilon}{\ln a} \rightarrow \infty \quad \text{as} \quad a \rightarrow 1$$

The one-percent time constant is of course $n_{\text{eff}} = \frac{\ln(0.01)}{\ln(a)}$

The 3-DB cutoff frequency ω_c is calculated at $|H(\omega_c)|^2 = \frac{1}{2}$, leading to

$$\omega_c = 1 - \frac{(1 - a)^2}{2a} \underset{\text{for } a \lesssim 1}{\approx} 1 - a$$

Highpass signal extraction filter S389 The NRR, n_{eff} and ω_c for a highpass IIR filter with

$$H(z) = \frac{b}{1 + a z^{-1}}$$

can be calculated similarly, leading to the same results:

$$\text{NRR} = \frac{1 - a}{1 + a} \quad n_{\text{eff}} = \frac{\ln \epsilon}{\ln a} \quad \omega_c \lesssim 1 - a$$

First-order IIR smoother with prescribed cutoff frequency S390 By adding a zero at $z = -1$, the NRR can be slightly improved. The filter will be

$$H(z) = \frac{b(1 + z^{-1})}{1 - a z^{-1}}$$

with $b = \frac{1-a}{2}$ for unity gain at DC. The NRR of this filter will be

$$\text{NRR} = \frac{1 - a}{2}$$

The following equations are used to determine ω_c or a .

$$\cos \omega_c = \frac{2a}{1 + a^2} \Leftrightarrow a = \frac{1 - \tan(\omega_c/2)}{1 + \tan(\omega_c/2)}$$

FIR averaging filters S391 For a FIR filter, the NRR will always be

$$\text{NRR} = \sum_n h_n^2$$

A filter minimizing the NRR is obtained by setting $h_n = \frac{1}{N}$, which is a simple averaging filter. The NRR will be

$$\text{NRR} = \frac{1}{N}$$

The length N of a FIR filter should be chosen like the time constant n_{eff} for an IIR filter: $N = n_{\text{eff}} = \frac{\ln \epsilon}{\ln a}$

The cutoff frequency of an FIR averaging filter is

$$\omega_c = \frac{\pi}{N}$$

Highpass FIR signal extraction S396 The lowpass filter is changed to a highpass filter by substituting $z \rightarrow -z$, changing the impulse response to $h_n \rightarrow (-1)^n h_n$, leading to a highpass FIR filter with the impulse response

$$h_n = (-1)^n \frac{1}{N}$$

The noise reduction ratio stays the same.

Bandpass signal extraction S397 A FIR bandpass filter is achieved by a resonator filter with poles at $z = Re^{\pm j\omega_0}$.

$$H(z) = \frac{G}{1 + a_1 z^{-1} + a_2 z^{-2}}$$

$$h_n = \frac{G}{\sin \omega_0} R^n \sin(\omega_0 n + \omega_0) u(n)$$

where $a_1 = -2R \cos \omega_0$ and $a_2 = R^2$.

The NRR of this filter will be

$$\text{NRR} = \frac{(1 - R)(1 + R^2)(1 - 2R \cos(2\omega_0) + R^2)}{(1 + R)(1 - 2R^2 \cos(2\omega_0) + R^4)}$$

The recovered sinusoid will be shifted by the phase delay d of

$$d(\omega_0) = -\frac{\arg H(\omega_0)}{\omega_0}$$

2.2.2. Notch and Comb Filters S398-406

Two special cases of the signal enhancement/noise reduction problem arise when:

1. The noise signal $\nu(n)$ is periodic. The ideal filter will then be a notch filter.

2. The desired signal $s(n)$ is periodic. The ideal filter will then be a comb filter.

Usually all the harmonics of the fundamental frequency f_1 must be canceled. The sampling rate f_s is then chosen to be a multiple f_1 :

$$f_s = Df_1$$

The noise harmonics then occur at the D^{th} roots-of-unit frequencies:

$$f_k = k \frac{f_s}{D}$$

Notch filters The notch filter is then given by

$$H_{\text{notch}}(z) = b \frac{1 - z^{-D}}{1 - a z^{-D}}$$

with $b = \frac{1+a}{2}$ and $a = \rho^D$. Again a must be in the interval $0 \leq a < 1$.

For a given $\Delta\omega$ of the notch dips, a and b are obtained by

$$\beta = \tan\left(\frac{D\Delta\omega}{4}\right) \quad a = \frac{1 - \beta}{1 + \beta} \quad b = \frac{1}{1 + \beta}$$

Comb filters As a multi-notch filter can be obtained from a single notch filter using the substitution $z \rightarrow z^D$, which creates D replicas in the Nyquist band, any narrow lowpass filter can be transformed into a comb filter.

$$H(z) = b \frac{1 + z^{-1}}{1 - a z^{-1}} \rightarrow H_{\text{comb}}(z) = b \frac{1 + z^{-D}}{1 - a z^{-D}}$$

Given a 3-dB width for the peaks, the parameters can be calculated by

$$\beta = \tan\left(\frac{D\Delta\omega}{4}\right) \quad a = \frac{1 - \beta}{1 + \beta} \quad b = \frac{\beta}{1 + \beta}$$

Note that the presented notch and comb filters are complementary, i.e.

$$|H_{\text{comb}}(\omega)|^2 + |H_{\text{notch}}(\omega)|^2 = 1$$

The NRRs are the same as the NRRs for the simple smoothers, as the transform $z \rightarrow z^D$ simply inserts $D-1$ zeros in the original impulse response. Hence the sum does not change and the NRRs don't change.

Transfer Function from the Difference Equation The transfer function is obtained by taking the z -transform of the difference equation assuming zero initial conditions.

Example:

$$y(n) = x(n) + x(n - D)$$

gives

$$H(z) = 1 + z^{-D}.$$

Similarly,

$$y(n) = x(n) - x(n - D)$$

results in

$$H(z) = 1 - z^{-D}.$$

Quality Factor The quality factor Q describes how narrow the notch is.

$$Q = \frac{\omega_0}{\Delta\omega}$$

or equivalently

$$\Delta\omega = \frac{\omega_0}{Q}$$

A larger Q results in a narrower notch and higher frequency selectivity.

For periodic comb filters, the parameter a can be calculated from the quality factor as

$$a = \frac{1 - \tan\left(\frac{\pi}{2Q}\right)}{1 + \tan\left(\frac{\pi}{2Q}\right)}$$

A larger value of a produces a narrower comb response.

2.2.3. Signal Averaging **S421-427**

Signal averaging is equivalent to comb filtering, but with an FIR filter instead of an IIR one. It is derived by applying the transform $z \rightarrow z^D$ to the FIR averaging filter of length N :

$$\begin{aligned} H(z) &= \frac{1}{N} \left(1 + z^{-D} + z^{-2D} + \dots + z^{-(N-1)D} \right) \\ &= \frac{1}{N} \frac{1 - z^{-ND}}{1 - z^{-D}} \end{aligned}$$

Again, the NRR stays the same: $\text{NRR} = \frac{1}{N}$.

In the time domain, the filter is given by the following I/O difference equation:

$$y(n) = \frac{1}{N} [x(n) + x(n-D) + \dots + x(n-(N-1)D)]$$

As the IIR comb filter is more efficient, it is mostly used for real time processing. The FIR averaging filter is used when measuring a finite number N of periods of the same noisy signal.

For N periods of D samples, the total length will be ND . The i th period of the signal will then be

$$x_i(n) = x(iD + n)$$

where $i = 0, 1, \dots, N-1$ for a total of N periods.

The output will then be

$$\hat{y}(n) = \frac{1}{N} \sum_{i=0}^{N-1} x_i(n)$$

and the noise will, of course, be reduced by a factor of the NRR, i.e.

$$\sigma_v^2 = \frac{1}{N} \sigma_v^2$$

Circular Buffer Implementation Instead of shifting all delay elements, a circular buffer updates only one memory location per sample.

The circular pointer is updated according to

$$q = (q + 1) \bmod (D + 1)$$

The delayed sample is accessed by

$$q_D = (q + D) \bmod (D + 1).$$

The recursion becomes

$$w[q] = w[q_D] + x$$

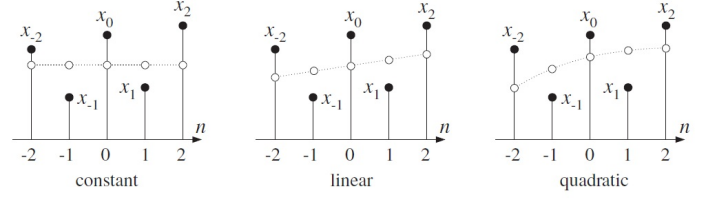
and the averaged output is

$$y = \frac{w[q]}{N}.$$

2.2.4. Savitzky-Golay Smoothing Filters S427-451

The Savitzky-Golay smoothing filters (also: polynomial smoothing or least-squares smoothing filters) are an elegant generalization of the simple FIR averaging filter.

It optimally fits a set of data points to a polynomial of a specified degree in least-squares sense.



The smoothed value $\hat{x}_m$ at each point is given by

$$\hat{x}_m = c_0 \quad \hat{x}_m = c_0 + c_1 m \quad \hat{x}_m = c_0 + c_1 m + c_2 m^2$$

with the free variable $m = -2, -1, 0, 1, 2$.

The coefficients c_i now have to be optimally found in a least-squares sense, using linear algebra.

We therefore introduce *basis vectors* $\mathbf{s}_0, \mathbf{s}_1, \mathbf{s}_2$, whose components are (quadratic approx.):

$$\mathbf{s}_0(m) = 1 \quad \mathbf{s}_1(m) = m \quad \mathbf{s}_2(m) = m^2$$

Therefore

$$\hat{\mathbf{x}} = [\mathbf{s}_0, \mathbf{s}_1, \mathbf{s}_2] \begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix} = S \mathbf{c}$$

with the optimal coefficients

$$\mathbf{c} = G^T \mathbf{x} \quad \text{with} \quad G = S (S^T S)^{-1}$$

Inserting the optimal coefficients, we find

$$\hat{\mathbf{x}} = SG^T \mathbf{x} = B \mathbf{x} \quad \text{with} \quad B = SG^T = S (S^T S)^{-1} S^T$$

And therefore the output smoothed value

$$y_0 = \hat{x}_0 = c_0 = \mathbf{b}_0^T \mathbf{x} = \mathbf{g}_0^T \mathbf{x}$$

This can be realized as a regular FIR filter, which is in case of the 2nd order polynomial

$$y_0 = \frac{1}{35} (-3x_{n-2} + 12x_{n-1} + 17x_n + 12x_{n+1} - 3x_{n+2})$$

The NRR of Savitzky-Golay filters is the middle coefficient:

$$\text{NRR} = \mathbf{b}_0^T \mathbf{b}_0 = b_0(0) = \frac{17}{35} = 0.49$$

Savitzky-Golay for derivatives Focus on the middle value at $m = 0$:

$$\hat{x}_m = c_0 + c_1 m + c_2 m^2$$

Clearly the derivatives are

$$\begin{aligned} \dot{\hat{x}}_0 &= \left. \frac{d\hat{x}_m}{dm} \right|_0 = c_1 = \mathbf{g}_1^T \mathbf{x} \\ \ddot{\hat{x}}_0 &= \left. \frac{d^2\hat{x}_m}{dm^2} \right|_0 = 2c_2 = 2\mathbf{g}_2^T \mathbf{x} \end{aligned}$$

and can be written as FIR filters

$$\begin{aligned}\dot{y}_n &= \frac{1}{35} (-7x_{n-2} - 3.5x_{n-1} + 3.5x_{n+1} + 7x_{n+2}) \\ \ddot{y}_n &= \frac{2}{35} (5x_{n-2} - 2.5x_{n-1} - 5x_n - 2.5x_{n+1} + 5x_{n+2})\end{aligned}$$

Savitzky-Golay for integrals The same concepts of course also apply to the integral i :

$$\begin{aligned}i_n &= \int_{-1/2}^{1/2} \hat{x}_m dm = \int_{-1/2}^{1/2} (c_0 + c_1 m + c_2 m^2) dm \\ &= [1 \ 0 \ 1/12] \cdot \mathbf{c} = [1 \ 0 \ 1/12] (S^T S)^{-1} S^T \mathbf{x}\end{aligned}$$

which leads to

$$\begin{aligned}\dot{y} &= (-0.0738x_{n-2} + 0.3369x_{n-1} + \\ &\quad 0.4738x_n + 0.3369x_{n+1} - 0.0738x_{n+2})\end{aligned}$$

Generalization So far, the length was $N = 5$ and the degree $d = 2$. Of course this can easily be generalized to any odd $N \geq d + 1$.

$$\begin{aligned}N &= 2M + 1 \\ \mathbf{x} &= [x_{-M}, \dots, x_{-1}, x_0, x_1, \dots, x_M]^T \\ \hat{x}_m &= c_0 + c_1 m + \dots + c_d m^d, \quad -M \leq m \leq M\end{aligned}$$

3. DFT/FFT Algorithms Chapter 9

3.1. Notation und verwendete Symbole

In der digitalen Signalverarbeitung werden in allen Kapiteln dieselben Symbole verwendet. Die wichtigsten Größen sind im Folgenden zusammengefasst.

Symbol	Bedeutung
f	Frequenz eines Signals in Hertz (Hz).
f_s	Abtastfrequenz (Sampling Frequency). Gibt an, wie viele Samples pro Sekunde aufgenommen werden. Einheit: Hz.
$T = \frac{1}{f_s}$	Abtastperiode (Sampling Period). Zeit zwischen zwei aufeinanderfolgenden Samples. Einheit: Sekunden.
f_0	Grundfrequenz bzw. gewünschte Frequenz eines Sinussignals.
ω	Digitale Kreisfrequenz in Radiant pro Sample. Es gilt $\omega = 2\pi \frac{f}{f_s}$ und der Nyquistbereich ist $-\pi \leq \omega \leq \pi$.
ω_0	Digitale Kreisfrequenz der Grundfrequenz f_0 : $\omega_0 = 2\pi \frac{f_0}{f_s}$.
T_D	Dauer einer Periode eines periodischen Signals mit $T_D = \frac{1}{f}$.
D	Anzahl Samples innerhalb einer Periode eines periodischen Signals. Oft auch Interpolations- oder Dezimationsfaktor.
L	Interpolationsfaktor. Zwischen zwei Samples werden $L-1$ neue Samples eingefügt. Die neue Abtastfrequenz beträgt $f'_s = L f_s$.
M	Dezimationsfaktor. Es wird nur jedes M -te Sample übernommen. Die neue Abtastfrequenz beträgt $f'_s = \frac{f_s}{M}$.

Symbol	Bedeutung
n	Diskreter Zeitindex ($n = 0, 1, 2, \dots$).
N	Anzahl Samples eines Signals oder Länge einer DFT bzw. FFT.
$x[n]$	Eingangssignal im Zeitbereich.
$y[n]$	Ausgangssignal im Zeitbereich.
$h[n]$	Impulsantwort eines Filters.
$H(z)$	Übertragungsfunktion eines digitalen Filters im z -Bereich.
$X[k]$	DFT bzw. FFT des Signals.
R	Radius der Pole im z -Diagramm. Für $0 < R < 1$ ist das System stabil. Je näher R an 1 liegt, desto langsamer klingt das Signal ab.
z^{-1}	Verzögerung um genau ein Sample (Unit Delay).

Wichtige Zusammenhänge

$$T = \frac{1}{f_s}$$

$$f = \frac{1}{T_D}$$

$$D = \frac{T_D}{T}$$

$$D = \frac{f_s}{f}$$

$$D = \frac{f_s}{f}$$

$$f = \frac{f_s}{D}$$

$$\omega = 2\pi \frac{f}{f_s}$$

$$\omega_0 = 2\pi \frac{f_0}{f_s}$$

$$N = f_s T$$

bzw.

$$N = f_s \cdot t_{\text{signal}}$$

3.2. Basics

DTFT Discrete-time Fourier transform

Discrete time-domain, continuous frequency-domain.

DFT Discrete Fourier transform

Both time-domain and frequency-domain are discrete.

FFT Fast Fourier transform

A fast implementation of the DFT.

3.3. Frequency Resolution and Windowing S464-472

Given an infinite sampled sequence $x(nT)$, to calculate its spectrum a finite number of samples L has to be cut out using a window function $w(n)$. At the sampling time interval $T = 1/f_s$, the duration of the data record is $T_L = LT$.

The finite signal x_L is achieved by

$$x_L(n) = x(n)w(n)$$

using one of the below defined window functions.

Rectangular Window

$$w(n) = \begin{cases} 1, & \text{if } 0 \leq n \leq L-1 \\ 0, & \text{otherwise} \end{cases}$$

Hamming Window

$$w(n) = \begin{cases} 0.54 - 0.46 \cos\left(\frac{2\pi n}{L-1}\right), & \text{if } 0 \leq n \leq L-1 \\ 0, & \text{otherwise} \end{cases}$$

The constant c , which depends on the window, is introduced. For the *rectangular* window, $c = 1$, for the *Hamming* window, $c = 2$. c is always greater than one.

The mainlobe width of a given window is defined as

$$\Delta\omega_w = c \cdot \frac{2\pi}{L} \quad \text{or} \quad \Delta f_w = c \cdot \frac{1}{T_L}$$

The relative sidelobe level is

$$R = 20 \log_{10} \left| \frac{W(\omega)}{W(0)} \right|$$

at $\omega =$ position of the first sidelobe. ($\omega = 3\pi/L$ for the rectangular window). This leads to $R = -13.46\text{dB}$ for the rectangular and $R = -40\text{dB}$ for the Hamming window.

The frequency resolution, i.e. the condition that two sinusoids appear as two distinct ones is then

$$\Delta\omega \geq \Delta\omega_w = c \cdot \frac{2\pi}{L} \quad \text{or} \quad \Delta f \geq \Delta f_w = c \cdot \frac{f_s}{L}$$

For a given frequency resolution, the window length is therefore

$$L \geq c \cdot \frac{f_s}{\Delta f} = c \cdot \frac{2\pi}{\Delta\omega}$$

Einfluss des Fensters Ein Hann-Fenster reduziert Leakage, verringert jedoch die Amplitude der Spektrallinien und verbreitert die Hauptkeule.

Merke Die physikalische Frequenzauflösung wird durch die Messdauer bestimmt:

$$\Delta f = \frac{1}{T} = \frac{f_s}{L}$$

Die DFT besitzt dagegen eine Binbreite

$$\Delta f_{\text{bin}} = \frac{f_s}{N}.$$

Nur eine längere Messdauer verbessert die tatsächliche Frequenzauflösung. Zero Padding erhöht lediglich die Anzahl der DFT-Punkte.

3.4. DTFT Computation S475-482**3.4.1. DTFT at a Single Frequency S475-477**

$$\begin{aligned} X(\omega) &= \sum_{n=0}^{L-1} x(n)e^{-j\omega n} = \sum_{n=0}^{L-1} x(n)z^{-n} \Big|_{z=e^{j\omega}} \\ &= X(z)|_{z=e^{j\omega}} \end{aligned}$$

This polynomial can be evaluated using *Hörner's rule*:

for each complex z do:

$X = 0$

for $n = L-1$ down to $n = 0$ do:

$X = x_n + z^{-1}X$

for example, let $L = 4$:

$X = 0$

$X = x_3 + z^{-1}X = x_3$

$X = x_2 + z^{-1}X = x_2 + z^{-1}x_3$

$X = x_1 + z^{-1}X = x_1 + z^{-1}x_2 + z^{-2}x_3$

$X = x_0 + z^{-1}X = x_0 + z^{-1}x_1 + z^{-2}x_2 + z^{-3}x_3 = X(z)$

3.4.2. DFT S479-481

The N -point DFT of a length- L signal is defined as the DTFT evaluated at N equally spaced frequencies over the full Nyquist interval, $0 \leq \omega \leq 2\pi$. These frequencies are

$$\omega_k = \frac{2\pi k}{N}, \quad f_k = \frac{k f_s}{N}, \quad k = 0, 1, \dots, N-1$$

Thus, the DFT is

$$X(\omega_k) = \sum_{n=0}^{L-1} x(n)e^{-j\omega_k n} = \sum_{n=0}^{L-1} x(n)z_k^{-n}$$

where z_k are the N th roots of unity

$$z_k = e^{j\omega_k} = e^{2\pi j k / N}$$

which leads to

$$\omega_{k+N} = \omega_k + 2\pi \quad \text{and} \quad X(k+N) = X(k)$$

The bin width is then

$$\Delta\omega_{\text{bin}} = \frac{2\pi}{N} \quad \text{or} \quad \Delta f_{\text{bin}} = \frac{f_s}{N}$$

3.4.3. Zero Padding S481

Note that L is the number of time moments and N is the number of frequency points. Mostly $L = N$ is used. For $L < N$, zero padding can be used to make the time sequence longer. For $L > N$, a modulo- N wrapping can be used to make the time sequence shorter.

Adding any number of zeros to the end of a signal does *not* change its DTFT!

$$\begin{aligned}\mathbf{x} &= [x_0, x_1, \dots, x_{L-1}] \\ \mathbf{x}_D &= [x_0, x_1, \dots, x_{L-1}, \underbrace{0, 0, \dots, 0}_{D \text{ zeros}}] \\ X_D(\omega_k) &= X(\omega_k)\end{aligned}$$

Adding zeros in front adds a linear phase term.

$$\begin{aligned}\mathbf{x} &= [x_0, x_1, \dots, x_{L-1}] \\ \mathbf{x}_D &= [\underbrace{0, 0, \dots, 0}_{D \text{ zeros}}, x_0, x_1, \dots, x_{L-1}] \\ X_D(\omega_k) &= e^{-j\omega_k D} X(\omega_k)\end{aligned}$$

3.5. Physical vs Computational Resolution S482-485

Physical frequency resolution is defined as

$$\Delta f_w = c \cdot \frac{f_s}{L}$$

whereas the *computational* frequency resolution is

$$\Delta f_{\text{bin}} = \frac{f_s}{N}$$

The DFT index k is proportional to the frequency f via

$$k = N \frac{f}{f_s}, \quad 0 \leq k \leq N \text{ and } 0 \leq f \leq f_s$$

Ideally, each frequency f_0 in the original signal coincides with one of the N DFT frequencies $f_s k/N$. This is so if k_0 is an integer.

$$k_0 = N \frac{f_0}{f_s}$$

Usually, this is not the case and the DFT will miss the peak. If N is large enough, the *rounding error* $= f_s/2N$ will stay small enough.

The *biasing error*, on the other hand, can only be decreased by increasing the data length L , because the DFT is only a sampled version of the DTFT.

Known Sinusoids If

$$x(n) = A \cos\left(2\pi \frac{k_0}{N} n\right),$$

then only two FFT coefficients are nonzero:

$$k = k_0$$

and

$$k = N - k_0.$$

Each coefficient has magnitude

$$\frac{AN}{2}.$$

A DC component appears only at

$$k = 0.$$

Merke Ein Sinus mit der Frequenz f_0 erscheint im DFT-Bin

$$k = \frac{N f_0}{f_s}.$$

Für reelle Signale entsteht zusätzlich der Spiegelpeak bei

$$N - k.$$

Liegt k nicht auf einer ganzen Zahl, verteilt sich die Energie auf mehrere Bins (Leakage).

3.6. Interpretation der FFT

Für ein reelles Signal gilt:

$$k = \frac{N f_0}{f_s}$$

Der FFT-Bin k entspricht der Frequenz

$$f_k = \frac{k f_s}{N}.$$

Ein Cosinus erscheint immer als zwei Spektrallinien:

$$k \quad \text{und} \quad N - k.$$

Ein Sinus besitzt dieselben FFT-Indizes, jedoch rein imaginäre Spektralwerte.

Merke

$\cos(\omega_0 n) \implies$ reelle Peaks

$\sin(\omega_0 n) \implies$ imaginäre Peaks

DC-Anteil:

$$k = 0$$

Nyquist-Frequenz:

$$k = \frac{N}{2} \quad (N \text{ gerade})$$

3.7. Matrix Form of DFT S486-488

The DFT can be defined as a linear matrix transformation

$$\mathbf{X} = \text{DFT}(\mathbf{x}) = A\mathbf{x}$$

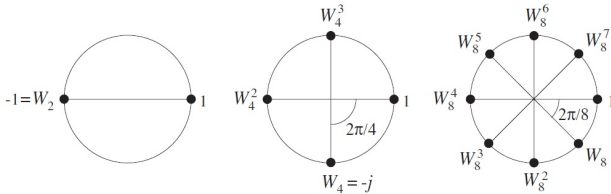
The matrix A for a N -point DFT can be built using the so-called *twiddle factor*.

$$W_N = e^{-2\pi j/N}$$

The matrix elements A_{kn} are found using the definition

$$A_{kn} = W_N^{kn}$$

The powers of the twiddle factors are the N -th roots of unity:



Some examples of twiddle factors and DFT matrices (see appendix A for more examples):

$$W_2 = e^{-2\pi j/2} = -1$$

$$W_4 = e^{-2\pi j/4} = \cos(\pi/2) - j \sin(\pi/2) = -j$$

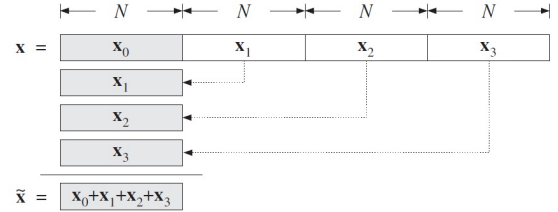
$$W_8 = e^{-2\pi j/8} = \cos(\pi/4) - j \sin(\pi/4) = \frac{1-j}{\sqrt{2}}$$

$$A_2 = \begin{bmatrix} 1 & 1 \\ 1 & W_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$A_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & W_4 & W_4^2 & W_4^3 \\ 1 & W_4^2 & W_4^4 & W_4^6 \\ 1 & W_4^3 & W_4^6 & W_4^9 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

3.8. Modulo-N reduction S489-496

Modulo-N reduction is applied if $L > N$. The input signal is divided into non-overlapping blocks of length N and added up.



$$\tilde{\mathbf{x}} = \mathbf{x}_0 + \mathbf{x}_1 + \mathbf{x}_2 + \mathbf{x}_3 \quad \text{or} \quad \tilde{x}(n) = \sum_{m=0}^{\infty} x(mN + n)$$

The proof that $\mathbf{X} = \tilde{\mathbf{X}}$ is that

$$\mathbf{X} = A\mathbf{x} = [\tilde{A}, \tilde{A}, \dots, \tilde{A}] \begin{bmatrix} \mathbf{x}_0 \\ \mathbf{x}_1 \\ \vdots \\ \mathbf{x}_n \end{bmatrix} = \tilde{A}\mathbf{x}_0 + \tilde{A}\mathbf{x}_1 + \dots$$

$$= \tilde{A}(\mathbf{x}_0 + \mathbf{x}_1 + \dots + \mathbf{x}_n) = \tilde{A}\tilde{\mathbf{x}} = \tilde{\mathbf{X}}$$

Similarly, a Matrix A can be created to transform e.g. a length-8 time signal into a 4-point DFT:

$$A = [\tilde{A}, \tilde{A}] = \left[\begin{array}{cccc|cccc} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & W_4 & W_4^2 & W_4^3 & 1 & W_4 & W_4^2 & W_4^3 \\ 1 & W_4^2 & W_4^4 & W_4^6 & 1 & W_4^2 & W_4^4 & W_4^6 \\ 1 & W_4^3 & W_4^6 & W_4^9 & 1 & W_4^3 & W_4^6 & W_4^9 \end{array} \right]$$

Signals with the Same DFT Signals that differ only by multiples of the DFT length produce the same DFT after modulo- N reduction.

Only the modulo- N wrapped sequence determines the DFT.

3.9. Inverse DFT S496-499

The problem of inverting an N -point DFT is that all signals $\mathbf{x}$ with the same modulo- N reduction $\tilde{\mathbf{x}}$ have the same N -point DFT.

$$\mathbf{X} = A\mathbf{x} = \tilde{A}\tilde{\mathbf{x}}$$

which means that only the modulo- N reduction of the input signal $\mathbf{x}$ can be reconstructed. Thus

$$\begin{aligned} \tilde{\mathbf{x}} &= \mathbf{x} & \text{if } N \geq L \\ \tilde{\mathbf{x}} &\neq \mathbf{x} & \text{if } N < L \end{aligned}$$

The inverse DFT is thus defined as

$$\tilde{\mathbf{x}} = \text{IDFT}(\mathbf{X}) = \tilde{A}^{-1}\mathbf{X}$$

As $\frac{1}{N}\tilde{A}\tilde{A}^* = I_N$, the inverse of the matrix $\tilde{A}$ can be obtained by $\tilde{A}^{-1} = \frac{1}{N}\tilde{A}^*$, which leads to the definition

$$\tilde{\mathbf{x}} = \text{IDFT}(\mathbf{X}) = \frac{1}{N}\tilde{A}^*\mathbf{X} = \frac{1}{N}[\text{DFT}(\mathbf{X}^*)]^*$$

Merke Die IFFT liefert wieder das Zeitsignal.
Reelle Spektren ergeben Cosinusanteile.
Rein imaginäre Spektren ergeben Sinusanteile.

3.10. FFT S504-515

The fast Fourier transform (FFT) is a fast implementation of the DFT. Not an approximation!

The FFT calculates the DFT using a divide-and-conquer strategy. It is based dividing the dimensions of the DFT in half until it becomes one. Hence the dimension N of the FFT is usually a power of 2: $N = 2^B$.

The computational cost can be reduced from N^2 for the DFT to $\frac{1}{2}N \log_2(N)$ for the FFT.

A length- N sequence $x(n)$ can be divided into two length- $(N/2)$ subsequences:

$$\begin{aligned} g(n) &= x(2n) \\ h(n) &= x(2n+1) \end{aligned}, \quad n = 0, 1, \dots, \frac{N}{2} - 1$$

with their corresponding DFTs

$$\begin{aligned} G(k) &= \sum_{n=0}^{N/2-1} W_{N/2}^{kn} g(n) \\ H(k) &= \sum_{n=0}^{N/2-1} W_{N/2}^{kn} h(n) \end{aligned}, \quad k = 0, 1, \dots, \frac{N}{2} - 1$$

The DFT of the original sequence $x(n)$ is obtained by

$$X(k) = G(k) + W_N^k H(k), \quad k = 0, 1, \dots, N-1$$

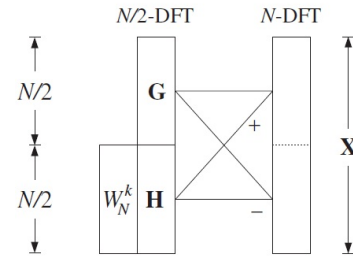
The *butterfly merging* equations follow from the periodicity properties of the DFT and the fact that the twiddle factor $W_N^{N/2} = -1$:

$$\begin{aligned} X(k) &= G(k) + W_N^k H(k) \\ X(k + N/2) &= G(k) - W_N^k H(k) \end{aligned} \quad \text{with } k = 0, 1, \dots, \frac{N}{2} - 1$$

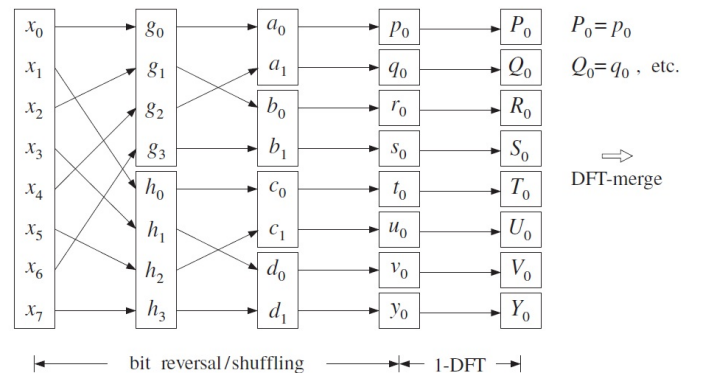
or more easily shown in a vector form:

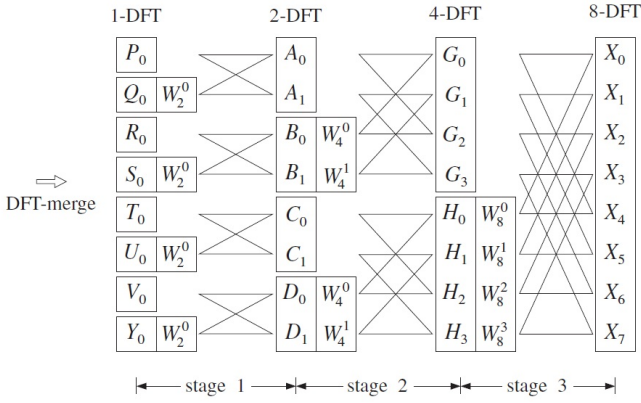
$$\begin{aligned} \begin{bmatrix} X_0 \\ X_1 \\ \vdots \\ X_{N/2-1} \end{bmatrix} &= \begin{bmatrix} G_0 \\ G_1 \\ \vdots \\ G_{N/2-1} \end{bmatrix} + \begin{bmatrix} H_0 \\ H_1 \\ \vdots \\ H_{N/2-1} \end{bmatrix} \times \begin{bmatrix} W_N^0 \\ W_N^1 \\ \vdots \\ W_N^{N/2-1} \end{bmatrix} \\ \begin{bmatrix} X_{N/2} \\ X_{N/2+1} \\ \vdots \\ X_{N-1} \end{bmatrix} &= \begin{bmatrix} G_0 \\ G_1 \\ \vdots \\ G_{N/2-1} \end{bmatrix} - \begin{bmatrix} H_0 \\ H_1 \\ \vdots \\ H_{N/2-1} \end{bmatrix} \times \begin{bmatrix} W_N^0 \\ W_N^1 \\ \vdots \\ W_N^{N/2-1} \end{bmatrix} \end{aligned}$$

or even graphically:

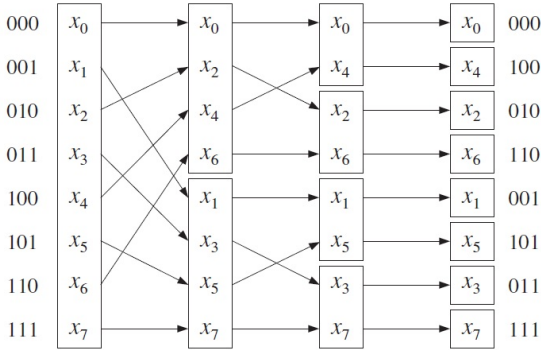


The FFT can easily be summarized by:





Note that the bit shuffling is also called bit reversal, because it has the same effect as binary reversing the input.



3.11. Fast Convolution S515-523

3.12. Overlap-Save Method

The overlap-save method performs long convolutions efficiently by using the FFT.

For a filter of length

$$M,$$

an FFT length

$$N > M$$

is selected.

Each input block contains

$$L = N - M + 1$$

new samples.

The remaining

$$M - 1$$

samples are copied from the previous block.

After multiplication in the frequency domain and the inverse FFT, the first

$$M - 1$$

output samples are discarded because they contain circular convolution artifacts.

Only the remaining

$$L$$

samples are valid.

$$L = N - M + 1$$

3.12.1. Circular Convolution S515-520

$$\mathbf{y} = \mathbf{h} * \mathbf{x} \quad \Leftrightarrow \quad Y(\omega) = H(\omega)X(\omega)$$

This can also be achieved by using the DFT:

$$\hat{\mathbf{y}} = \text{IDFT}(\text{DFT}(\mathbf{h}) \cdot \text{DFT}(\mathbf{x}))$$

But: the length L_y of the convolution of a length L and a length M input is $L_y = L + M$. To prevent a *circular convolution*, the input vectors have to be zero-padded to a length of

$$N \geq L_y = L + M$$

Relationship to Linear Convolution Circular convolution is obtained by reducing the linear convolution modulo N .

$$y_c(n) = y_{\text{linear}}(n) \bmod N.$$

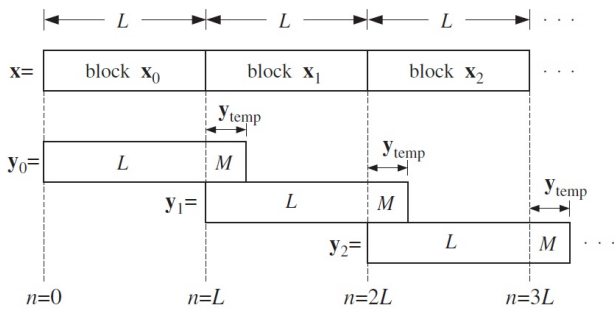
If both signals are zero-padded to at least

$$N \geq L_x + L_h - 1,$$

the circular convolution is identical to the linear convolution.

3.12.2. Overlap-Add and Overlap-Save Methods S520-523

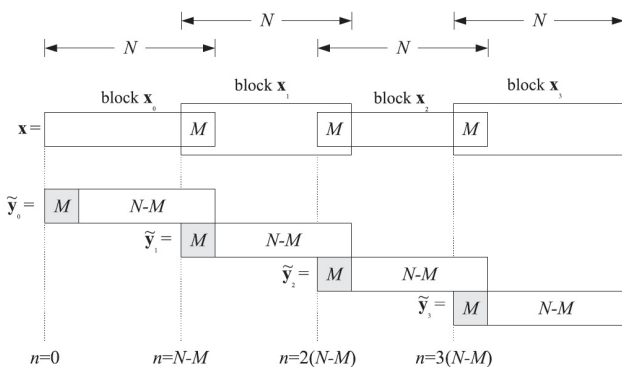
Overlap-Add If the input signal $\mathbf{x}$ is of infinite length, no zero-padding is possible. Then the input is divided into blocks $\mathbf{x}_0, \mathbf{x}_1, \dots$ which are processed separately. The FFT length N is of course $N \geq L + M$. As M is saved and added to the next result, no wraparound occurs.



The cost of the fast Overlap-Add method compared to the conventional, slow method is

$$\frac{\text{fast}}{\text{slow}} = \frac{N(\log_2 N + 1)}{(M + 1)(N - M)} \simeq \frac{\log_2 N}{M}$$

Overlap-Save Here, each block is of length $L = N$, but the blocks overlap by M . The output blocks will have M samples wrapped around, but as only the last $N - M$ samples are used, the M wrapped around samples are not used at all. (Though, the very first M samples will be incorrect.)



3.13. Rechenaufwand von DFT-Algorithmen

Für die Berechnung einer DFT gibt es verschiedene Algorithmen mit unterschiedlichem Rechenaufwand. Da Multiplikationen wesentlich aufwändiger sind als Additionen, wird meist die Anzahl der komplexen Multiplikationen verglichen.

Direkte DFT

Für jedes der N Ausgabesamples werden N Multiplikationen benötigt.

$$N_{\text{mult}} = N^2$$

Beispiel:

$$N = 16 \quad \Rightarrow \quad 16^2 = 256$$

DFT mit Modulo- N -Reduktion

Durch die Modulo- N -Eigenschaft werden identische Twiddle-Faktoren zusammengefasst. Dadurch reduziert sich die Anzahl der Multiplikationen.

$$N_{\text{mult}} = N(2N - 1)$$

(Falls in den Übungen nach der Anzahl der Twiddle-Faktor-Multiplikationen gefragt wird.)

FFT (Radix-2)

Bei der FFT wird das Problem rekursiv halbiert.
Anzahl Stufen:

$$\log_2 N$$

Butterflies pro Stufe:

$$\frac{N}{2}$$

Jeder Butterfly benötigt genau eine komplexe Multiplikation.

Damit ergibt sich

$$N_{\text{mult}} = \frac{N}{2} \log_2 N$$

Beispiel:

$$N = 16$$

$$\frac{16}{2} \log_2(16) = 8 \cdot 4 = 32$$

FFT mit Modulo- N -Reduktion

Durch Ausnutzen der Symmetrien der Twiddle-Faktoren können nochmals Multiplikationen eingespart werden.

Faustregel:

$$N_{\text{mult}} < \frac{N}{2} \log_2 N$$

Die genaue Anzahl hängt von den verwendeten Twiddle-Faktoren ab und wird meist direkt aus dem Butterfly-Diagramm bestimmt.

Zusammenfassung

Verfahren	Komplexe Multiplikationen
DFT	N^2
DFT (Modulo)	abhängig von der Modulo-Reduktion
FFT	$\frac{N}{2} \log_2 N$
FFT (Modulo)	$< \frac{N}{2} \log_2 N$

4. FIR Digital Filter Design Chapter 10

4.1. Window Method S532-541

4.1.1. Ideal Filters S532-534

The design of an FIR filter using the window method starts from an ideal frequency response $D(\omega)$. Since ideal filters are non-causal and have an infinite impulse response, they cannot be implemented directly. Therefore, the infinite impulse response is truncated by applying a window function.

The ideal frequency response and its corresponding impulse response are related by the inverse DTFT

$$d(k) = \int_{-\pi}^{\pi} D(\omega) e^{j\omega k} \frac{d\omega}{2\pi} \quad (1)$$

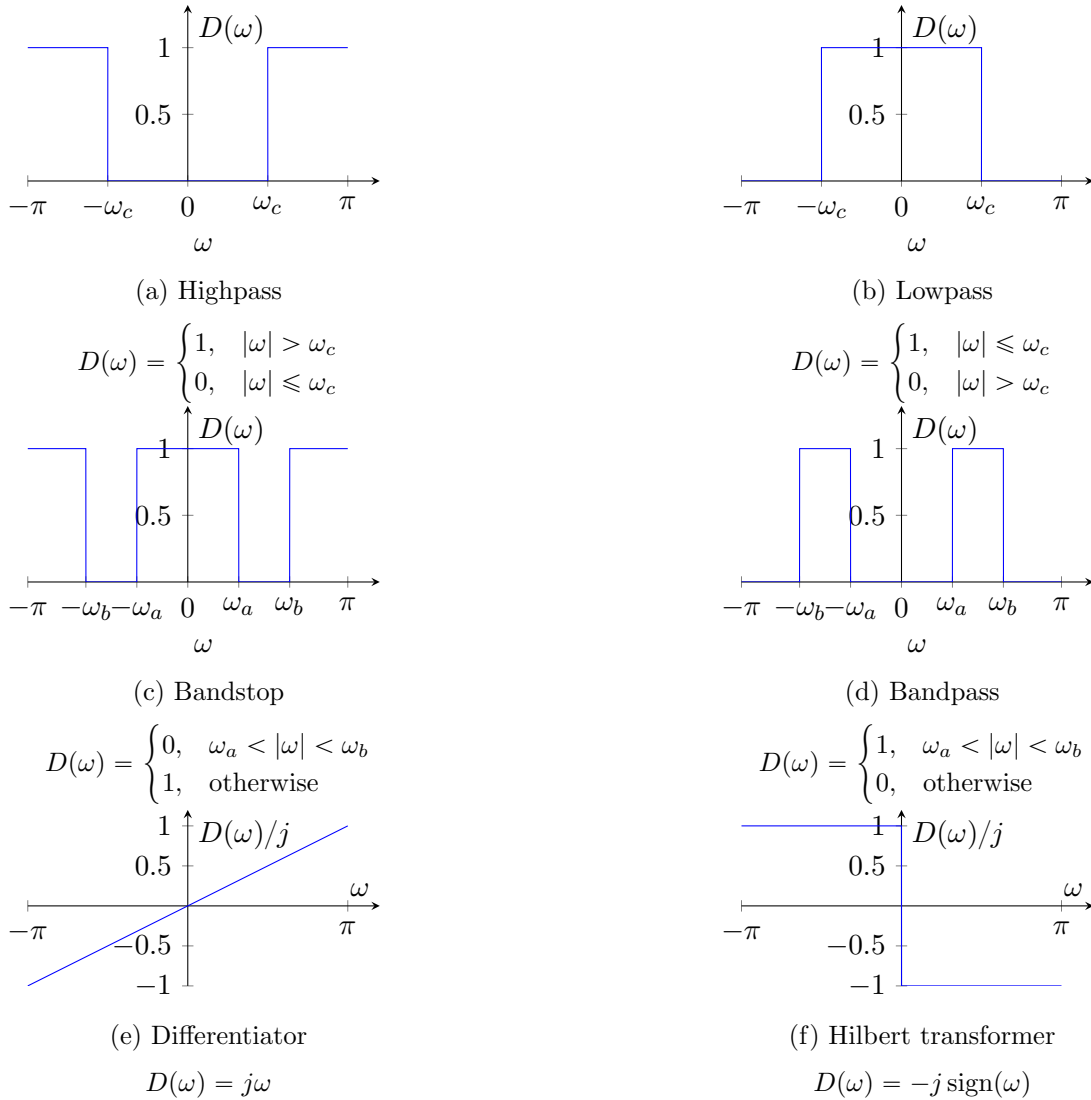


Figure 1: Ideal desired frequency responses $D(\omega)$ S533

For the most common ideal filters, the impulse responses are

Lowpass	$d(k) = \frac{\sin(\omega_c k)}{\pi k}$
Highpass	$d(k) = \delta(k) - \frac{\sin(\omega_c k)}{\pi k}$
Bandpass	$d(k) = \frac{\sin(\omega_b k) - \sin(\omega_a k)}{\pi k}$
Bandstop	$d(k) = \delta(k) - \frac{\sin(\omega_b k) - \sin(\omega_a k)}{\pi k}$
Differentiator	$d(k) = \frac{\cos(\pi k)}{k} - \frac{\sin(\pi k)}{\pi k^2}$
Hilbert transformer	$d(k) = \frac{1 - \cos(\pi k)}{\pi k}$

For

$$k = 0$$

the corresponding limits have to be evaluated separately.

4.1.2. FIR Filter Properties

Complementary Filters

$$\begin{aligned} d_{LP}(k) + d_{HP}(k) &= \delta(k) & \iff D_{LP}(\omega) + D_{HP}(\omega) &= 1 \\ d_{BP}(k) + d_{BS}(k) &= \delta(k) & \iff D_{BP}(\omega) + D_{BS}(\omega) &= 1 \end{aligned}$$

Symmetry

	Symmetric	Antisymmetric
Impulse response	$d(k) = d(-k)$	$d(k) = -d(-k)$
Examples	LP, HP, BP, BS	Differentiator, Hilbert
Frequency response	real, even	imaginary, odd
Phase	linear	linear

FIR Implementation

$$y(n) = \sum_{m=0}^{N-1} h(m)x(n-m)$$

$h(m)$ FIR coefficients
 $x(n)$ input
 $y(n)$ output
 N filter length

Group Delay For a linear-phase FIR filter

$$\tau_g = \frac{N-1}{2} \text{ samples}$$

or

$$t_g = \frac{N-1}{2f_s}$$

where N is the filter length and f_s the sampling frequency.

4.1.3. Window Comparison

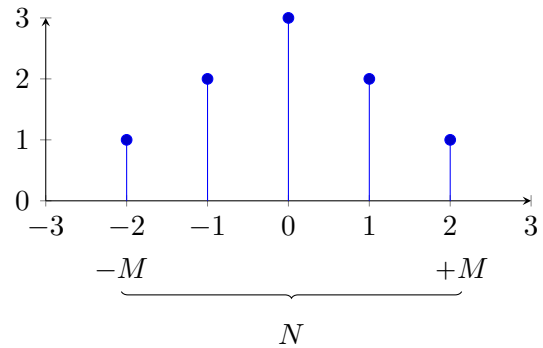
The most common window functions differ mainly in

- transition width,
- stopband attenuation,
- passband ripple,
- sidelobe attenuation.

Window	δ	A_{stop}	A_{pass}	D	R	c
Rectangular	8.9%	-21 dB	1.55 dB	0.922	-13 dB	1
Hamming	0.2%	-54 dB	0.03 dB	3.21	-40 dB	2
Kaiser	variable	$-20 \log_{10} \delta$	17.372δ	$\begin{cases} \frac{A-7.95}{14.36}, & A > 21 \\ 0.922, & A \leq 21 \end{cases}$	variable	$\frac{6(R+12)}{55}$

4.1.4. Rectangular Window S535

- Simplest window, Length $N = 2M + 1$
- truncate the infinite impulse response:
 $-M \leq k \leq M$
- $\rightarrow \mathbf{d} = [d_{-M}, \dots, d_{-1}, d_0, d_1, \dots, d_M]$
- The resulting causal FIR Filter must be delayed by M samples
 $\rightarrow \mathbf{h} = \mathbf{d} = [h_0, h_{M-1}, h_M, h_{M+1}, \dots, h_{2M}]$



Gibbs Phenomenon Truncating the infinite impulse response introduces oscillations close to the cutoff frequency. Characteristics:

- Overshoot approximately 8.9%
- Independent of the filter length
- Increasing N narrows the oscillation region
- The oscillation amplitude does not disappear

4.1.5. Hamming Window **S540**

- Improved passband and stopband ripple suppression at the cost of a wider transition width.
- Hamming window impulse response:

$$w(n) = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right), \quad n = 0, 1, \dots, N-1$$

- Filter impuls response:

$$h(n) = w(n)d(n-M), \quad -\infty < n < \infty$$

- e.g. lowpass filter

$$h(n) = \left[0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right) \right] \cdot \frac{\sin(\omega_c(n-M))}{\pi(n-M)}$$

Properties Compared to the rectangular window:

- much smaller passband ripple
- much smaller stopband ripple
- lower sidelobes
- wider transition band
- ripple approximately 0.2%

4.1.6. Hann Window

The Hann window is defined by

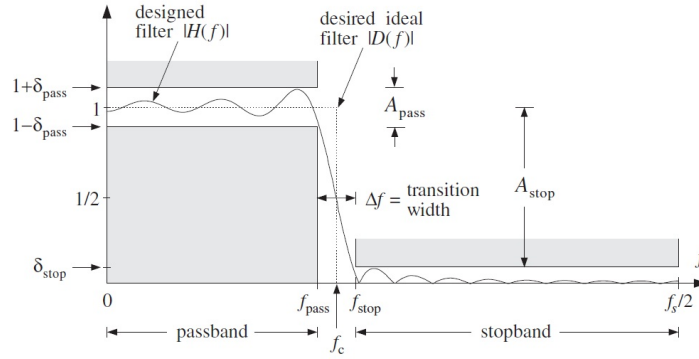
$$w(n) = 0.5 - 0.5 \cos\left(\frac{2\pi n}{N-1}\right), \quad n = 0, 1, \dots, N-1.$$

Compared to the Hamming window, the Hann window provides slightly wider main lobes but lower spectral leakage than the rectangular window.

4.1.7. Kaiser Window **S541**

The Kaiser window is a family of window functions whose shape can be adjusted by the parameter α . Unlike the rectangular and Hamming windows, the Kaiser window allows a trade-off between

- transition width,
- stopband attenuation,
- filter length.



Cutoff frequency	$f_c = \frac{1}{2}(f_{pass} + f_{stop})$
Transition width	$\Delta f = f_{stop} - f_{pass}$
Passband / Stopband frequencies	$f_{pass} = f_c - \frac{\Delta f}{2}, \quad f_{stop} = f_c + \frac{\Delta f}{2}$
Digital frequencies	$\omega_{pass} = \frac{2\pi f_{pass}}{f_s}, \quad \omega_{stop} = \frac{2\pi f_{stop}}{f_s}$ $\omega_c = \frac{2\pi f_c}{f_s}, \quad \Delta\omega = \frac{2\pi \Delta f}{f_s}$
Passband / Stopband ripple	$A_{pass} = 20 \log_{10} \left(\frac{1+\delta_{pass}}{1-\delta_{pass}} \right)$ $A_{stop} = -20 \log_{10}(\delta_{stop})$ $\delta_{pass} = \frac{10^{A_{pass}/20} - 1}{10^{A_{pass}/20} + 1}$ $\delta_{stop} = 10^{-A_{stop}/20}$
Window parameters	$\alpha = \begin{cases} 0.1102(A - 8.7), & A \geq 50 \\ 0.5842(A - 21)^{0.4} + 0.07886(A - 21), & 21 < A < 50 \\ 0, & A \leq 21 \end{cases}$ $D = \begin{cases} \frac{A - 7.95}{14.36}, & A > 21 \\ 0.922, & A \leq 21 \end{cases}$
Window function	$w(n) = \frac{I_0 \left(\alpha \sqrt{1 - \frac{(n-M)^2}{M^2}} \right)}{I_0(\alpha)}$ $= \frac{I_0 \left(\frac{\alpha \sqrt{n(2M-n)}}{M} \right)}{I_0(\alpha)}$ $I_0 = \text{modified Bessel function of first kind, order 0}$

Lowpass Design (Kaiser Window) S544

1. Cutoff and transition width

$$f_c = \frac{f_{pass} + f_{stop}}{2}, \quad \Delta f = f_{stop} - f_{pass}$$

2. Digital cutoff frequency

$$\omega_c = \frac{2\pi f_c}{f_s}$$

3. Passband and stopband ripples

$$\delta_{pass} = \frac{10^{A_{pass}/20} - 1}{10^{A_{pass}/20} + 1}, \quad \delta_{stop} = 10^{-A_{stop}/20}$$

4. Select the smaller ripple

$$\delta = \min(\delta_{pass}, \delta_{stop}), \quad A = -20 \log_{10} \delta$$

(Kaiser uses the smaller ripple.)

5. Calculate the Kaiser parameters

$$\alpha, \quad D$$

6. Filter length

$$N = \frac{Df_s}{\Delta f} + 1$$

Round N up to the next odd integer and calculate

$$M = \frac{N-1}{2}.$$

7. Kaiser window

$$w(n), \quad n = 0, \dots, N-1$$

8. FIR coefficients

$$h(n) = w(n)d(n-M)$$

4.1.8. Kaiser Design Equations

For a Kaiser window design the transition bandwidth and the filter length are approximately related by

$$N \approx \frac{D}{\Delta F}$$

where

$$\Delta F = \frac{\Delta f}{f_s}$$

is the normalized transition width.

The transition width is

$$\Delta f = f_{stop} - f_{pass}$$

The cutoff frequency is

$$f_c = \frac{f_{pass} + f_{stop}}{2}$$

or, in digital frequency,

$$\omega_c = 2\pi \frac{f_c}{f_s}.$$

4.1.9. Implementation Limits

For real-time DSP implementations, the maximum filter length is limited by the available instruction rate.

$$N_{\max} \approx \frac{f_{\text{instr}}}{f_s}$$

where

- f_{instr} is the DSP instruction rate,
- f_s is the sampling frequency.

The corresponding minimum transition bandwidth is

$$\Delta f_{\min} = \frac{D f_s^2}{f_{\text{instr}}}$$

where D is the Kaiser design parameter.

Maximum Achievable Stopband Attenuation For a fixed DSP instruction rate the maximum achievable attenuation is

$$A_{\max} = 14.36 F_{\text{instr}} \Delta F + 7.95$$

with

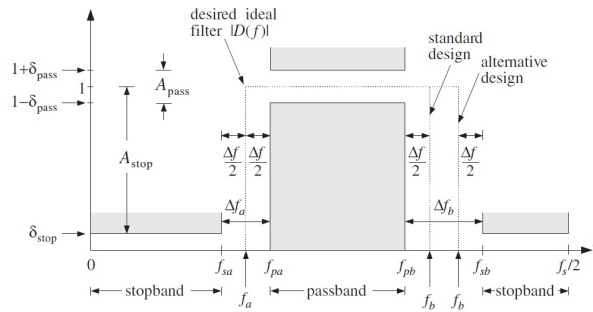
$$F_{\text{instr}} = \frac{f_{\text{instr}}}{f_s}, \quad \Delta F = \frac{\Delta f}{f_s}.$$

4.1.10. Rules for Different Filter Types

Highpass Filter For a highpass filter, the passband and stopband are exchanged. Therefore,

$$\Delta f = f_{\text{pass}} - f_{\text{stop}}.$$

Bandpass Filter



The final filter has equal transition widths

$$\Delta f = \min(\Delta f_a, \Delta f_b)$$

with

$$\Delta f_a = f_{pa} - f_{sa},$$

$$\Delta f_b = f_{sb} - f_{pb}.$$

Two design choices are possible.

Focus on Passband

$$f_a = f_{pa} - \frac{\Delta f}{2},$$

$$f_b = f_{pb} + \frac{\Delta f}{2}.$$

Focus on Stopband

$$f_a = f_{sa} + \frac{\Delta f}{2},$$

$$f_b = f_{sb} - \frac{\Delta f}{2}.$$

4.1.11. Kaiser Window for Spectral Analysis S555

The Kaiser window can also be used for spectral analysis (DFT and FFT).

For a desired sidelobe level

$$R$$

the parameter

$$c$$

is calculated as

$$c = \frac{6(R + 12)}{155}$$

The corresponding Kaiser shape parameter is

$$\alpha = \begin{cases} 0, & R < 13.26 \\ 0.76609(R - 13.26)^{0.4} + 0.09834(R - 13.26), & 13.26 < R < 60 \\ 0.12438(R + 6.3), & 60 < R < 120. \end{cases}$$

4.2. Frequency Sampling Method S558

For an arbitrary frequency response

$$D(\omega),$$

the inverse DTFT generally cannot be evaluated analytically. Therefore, the impulse response is approximated by sampling the desired frequency response at equally spaced frequencies.

Instead of

$$d(k) = \int_{-\pi}^{\pi} D(\omega) e^{j\omega k} \frac{d\omega}{2\pi},$$

the approximation

$$\tilde{d}(k) = \frac{1}{N} \sum_{i=-M}^M D(\omega_i) e^{j\omega_i k}$$

is used, where

$$-M \leq k \leq M.$$

If

$$N = 2M + 1,$$

the approximation corresponds to an inverse

$$N\text{-point DFT.}$$

Windowing As with the Window Method, the sampled impulse response is multiplied by a window function.

$$h(n) = w(n) \tilde{d}(n - M), \quad n = 0, 1, \dots, N - 1.$$

Therefore, every window introduced previously (Rectangular, Hamming, Kaiser, ...) can also be used together with the Frequency Sampling Method.

Design Procedure The Frequency Sampling Method consists of the following steps:

1. Specify the desired frequency response $D(\omega)$.
2. Sample the frequency response at equally spaced frequencies.
3. Perform an inverse DFT to obtain the impulse response.
4. Apply a suitable window function.
5. Shift the impulse response by $M = (N - 1)/2$ samples to obtain a causal FIR filter.

Comparison with the Window Method

Window Method	Frequency Sampling Method
Starts from an analytical ideal response	Starts from discrete frequency samples
Requires inverse DTFT	Requires inverse DFT
Simple for standard filters	Suitable for arbitrary frequency responses
Window applied after truncation	Window applied after IDFT

4.3. Other FIR Design Methods S558

Although the Kaiser Window Method is simple and flexible, other FIR design techniques often achieve a better approximation with shorter filters.
The most important alternative is the

$$\text{Parks-McClellan Algorithm}$$

which is based on the

$$\text{Equiripple Chebyshev Approximation.}$$

Characteristics Compared with the Window Method,

- the passband and stopband ripples have equal height,
- the approximation error is minimized,
- shorter filters are often sufficient,
- the design is computationally more involved.

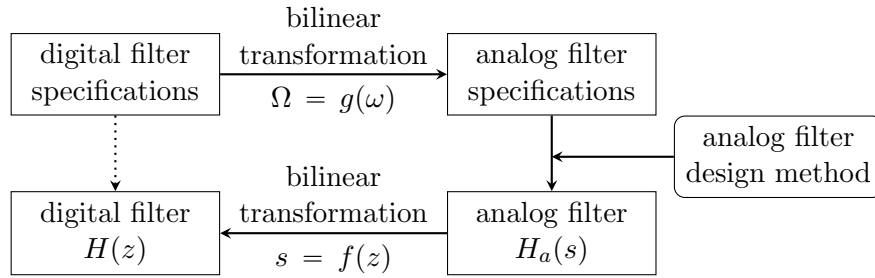
Comparison

Window Method	Parks–McClellan
Simple design	Optimal design
Longer filters	Shorter filters
Window dependent	Equiripple approximation
Very easy to implement	Iterative optimisation

5. IIR Digital Filter Design Chapter 11

5.1. Bilinear Transformation S563-565

Designflow:



Maps from z-plane to the s-plane (from the analog frequency to the digital frequency)

Filter	bilinear transformation	frequency map
lowpass	$s = f(z) = \frac{1-z^{-1}}{1+z^{-1}}$	$\Omega = g(\omega) = \tan\left(\frac{\omega}{2}\right)$
highpass	$s = f(z) = \frac{1+z^{-1}}{1-z^{-1}}$	$\Omega = g(\omega) = -\cot\left(\frac{\omega}{2}\right)$
bandpass	$s = f(z) = \frac{1-2cz^{-1}+z^{-2}}{1-z^{-2}}$	$\Omega = g(\omega) = \frac{c-\cos(\omega)}{\sin(\omega)}$
bandstop	$s = f(z) = \frac{1-z^{-2}}{1-2cz^{-1}+z^{-2}}$	$\Omega = g(\omega) = \frac{\sin(\omega)}{\cos(\omega)-c}$

5.2. First-Order LP/HP, Second-Order Peaking an Notching Filters S566-581

Definitions

cutoff/center frequency	f_c	
sampling rate	f_s	
bandwidth	Δf	Peaking/Notching filter only
digital cutoff/center frequency	ω_c	$= \frac{2\pi f_c}{f_s}$ (usually at 3dB-point)
digital center frequency	ω_0	$= \frac{2\pi f_0}{f_s}$
drop in dB at f_c	A_c	$= -10 \log_{10}(G_c^2) = -20 \log_{10}(G_c)$ (usually 3dB)
drop factor	G_c	$= 10^{-A_c/20} \longrightarrow \text{If } A_c = 3\text{dB} \rightarrow G_c^2 = 1/2$

filter	$H(z) =$	$H_a(s) =$	$\alpha =$	$a =$	$b =$
low-pass	$b \frac{1+z^{-1}}{1-az^{-1}}$	$\frac{\alpha}{s+\alpha}$	$\frac{G_c}{\sqrt{1-G_c^2}} \tan\left(\frac{\omega_c}{2}\right) = \tan\left(\frac{\omega_c}{2}\right) \big _{G_c^2=\frac{1}{2}}$	$\frac{1-\alpha}{1+\alpha}$	$\frac{\alpha}{1+\alpha} = \frac{1-a}{2}$
high-pass	$b \frac{1-z^{-1}}{1-az^{-1}}$	$\frac{s}{s+\alpha}$	$\frac{\sqrt{1-G_c^2}}{G_c} \tan\left(\frac{\omega_c}{2}\right) = \tan\left(\frac{\omega_c}{2}\right) \big _{G_c^2=\frac{1}{2}}$	$\frac{1-\alpha}{1+\alpha}$	$\frac{1}{1+\alpha} = \frac{1+a}{2}$
notch	$b \frac{1-2\cos(\omega_0)z^{-1}+z^{-2}}{1-2b\cos(\omega_0)z^{-1}+(2b-1)z^{-2}}$	$\frac{s^2+\Omega_0^2}{s^2+\alpha s+\Omega_0^2}$	$\frac{\sqrt{1-G_B^2}}{G_B} (1+\Omega_0^2) \tan\left(\frac{\Delta\omega}{2}\right)$	—	$\frac{1}{1+\frac{\sqrt{1-G_B^2}}{G_B} \tan\left(\frac{\Delta\omega}{2}\right)}$
peak	$(1-b) \frac{1-z^{-2}}{1-2b\cos(\omega_0)z^{-1}+(2b-1)z^{-2}}$	$\frac{\alpha s}{s^2+\alpha s+\Omega_0^2}$	$\frac{G_B}{\sqrt{1-G_B^2}} (1+\Omega_0^2) \tan\left(\frac{\Delta\omega}{2}\right)$	—	$\frac{1}{1+\frac{G_B}{\sqrt{1-G_B^2}} \tan\left(\frac{\Delta\omega}{2}\right)}$

Multi-Notch / Multi-Peak Filters Several notches or peaks are obtained by cascading individual sections.

$$H(z) = \prod_{k=1}^K H_k(z).$$

Each section has its own center frequency

$$\omega_{0,k} = \frac{2\pi f_{0,k}}{f_s}.$$

For a multi-notch filter,

$$H(z) = \prod_{k=1}^K \frac{1 - 2\cos(\omega_{0,k})z^{-1} + z^{-2}}{1 - 2b_k\cos(\omega_{0,k})z^{-1} + (2b_k - 1)z^{-2}}.$$

The same principle applies to multi-peak filters.

Poles and Zeros The poles are obtained by solving the denominator polynomial

$$D(z) = 0,$$

and the zeros by solving

$$N(z) = 0.$$

Example:

$$1 - 2\cos(\omega_0)z^{-1} + z^{-2} = 0.$$

Multiply by z^2 :

$$z^2 - 2\cos(\omega_0)z + 1 = 0.$$

Using the quadratic formula,

$$z_{1,2} = \cos(\omega_0) \pm j\sin(\omega_0) = e^{\pm j\omega_0}.$$

Hence the zeros lie on the unit circle at the angles

$$\boxed{z = e^{\pm j\omega_0}}.$$

Similarly,

$$1 - 2b\cos(\omega_0)z^{-1} + (2b - 1)z^{-2} = 0$$

gives the pole locations.

Complementary Filters The second-order notch and peaking filters are complementary.

$$\boxed{H_{notch}(z) + H_{peak}(z) = 1}$$

Both filters have identical poles and therefore identical transient behaviour.

Designflow

1. Given the cutoff frequency ω_c and the corresponding gain A_c in dB, compute G_c .
2. Compute the analog parameter α
3. Compute the digital filter coefficients $\{a, b\}$.

Pole Radius The poles of the second-order notch and peaking filters are located at

$$z = \rho e^{\pm j\omega_0},$$

where

$$\rho = \sqrt{\frac{1 - \beta}{1 + \beta}}$$

is the pole radius.

The pole radius determines the bandwidth and transient behaviour.

- $\rho \rightarrow 1$: narrow bandwidth, long transient.
- smaller ρ : wider bandwidth, short transient.

Effective Time Constant The transient response decays approximately as

$$\rho^n.$$

For an amplitude level

$$\varepsilon$$

the effective duration is

$$n_{\text{eff}} = \frac{\ln(\varepsilon)}{\ln(\rho)}$$

samples.

Typical values are

$$\varepsilon = 0.05$$

for the 5% settling time and

$$\varepsilon = 0.01$$

for the 1% settling time.

Small Bandwidth Approximation For narrowband filters

$$\Delta\omega \ll 1$$

the following approximations hold

$$\Delta\omega \approx 2(1 - \rho)$$

and therefore

$$n_{\text{eff}} \approx -\frac{2 \ln(\varepsilon)}{\Delta\omega}.$$

These approximations are useful for narrowband notch and peaking filters.

5.3. Higher Order Filters S592-631

The quantities $\{\epsilon_{\text{pass}}, \epsilon_{\text{stop}}\}$ control the depths of the passband and stopband. They can be written in the equivalent forms:

$$\begin{aligned} \epsilon_{\text{pass}} &= \sqrt{10^{A_{\text{pass}}/10} - 1} & \Longleftrightarrow & & A_{\text{pass}} &= 10 \log_{10}(1 + \epsilon_{\text{pass}}^2) \\ \epsilon_{\text{stop}} &= \sqrt{10^{A_{\text{stop}}/10} - 1} & \Longleftrightarrow & & A_{\text{stop}} &= 10 \log_{10}(1 + \epsilon_{\text{stop}}^2) \end{aligned}$$

The specifications of the equivalent analog filter are $\{\Omega_{\text{pass}}, \Omega_{\text{stop}}, A_{\text{pass}}, A_{\text{stop}}\}$, or, $\{\Omega_{\text{pass}}, \Omega_{\text{stop}}, \epsilon_{\text{pass}}, \epsilon_{\text{stop}}\}$, where the analog frequencies are obtained by pre-warping the digital frequencies.

$$\begin{aligned} \Omega_{\text{pass}} &= \tan\left(\frac{\omega_{\text{pass}}}{2}\right) & \Omega_{\text{stop}} &= \tan\left(\frac{\omega_{\text{stop}}}{2}\right) \\ \omega_{\text{pass}} &= \frac{2\pi f_{\text{pass}}}{f_s} & \omega_{\text{stop}} &= \frac{2\pi f_{\text{stop}}}{f_s} \end{aligned}$$

FIR vs. IIR Specifications FIR and IIR filters use different ripple parameters.

For FIR filters, the passband and stopband ripples are measured directly from the ideal magnitude response:

$$\begin{aligned} \delta_{\text{pass}} &= \frac{|H|_{\text{max}} - |H|_{\text{min}}}{|H|_{\text{max}} + |H|_{\text{min}}} \\ \delta_{\text{stop}} &= |H|_{\text{max, stop}}. \end{aligned}$$

For IIR filters, the specifications are expressed by

$$\begin{aligned} \epsilon_{\text{pass}} &= \sqrt{10^{A_{\text{pass}}/10} - 1} \\ \epsilon_{\text{stop}} &= \sqrt{10^{A_{\text{stop}}/10} - 1} \end{aligned}$$

because Butterworth and Chebyshev filters are naturally described by

$$|H(\Omega)|^2 = \frac{1}{1 + \epsilon^2 F(\Omega)}.$$

Thus,

$$\delta$$

is mainly used for FIR designs,
whereas

$$\epsilon$$

is used for IIR filter design.

5.3.1. Analog Lowpass Butterworth Filters S594-599

Analog lowpass Butterworth filters are characterized by just two parameters: filter order N and the 3-dB normalization frequency Ω_0 .

$$|H(\Omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{\Omega_0}\right)^{2N}}$$

With the given information, we can calculate the two parameters:

$$N_{exact} = \frac{\ln(\epsilon_{stop}/\epsilon_{pass})}{\ln(\Omega_{stop}/\Omega_{pass})} = \frac{\ln(e)}{\ln(w)} \quad e = \frac{\epsilon_{stop}}{\epsilon_{pass}} = \sqrt{\frac{10^{A_{stop}/10} - 1}{10^{A_{pass}/10} - 1}} \quad w = \frac{\Omega_{stop}}{\Omega_{pass}}$$

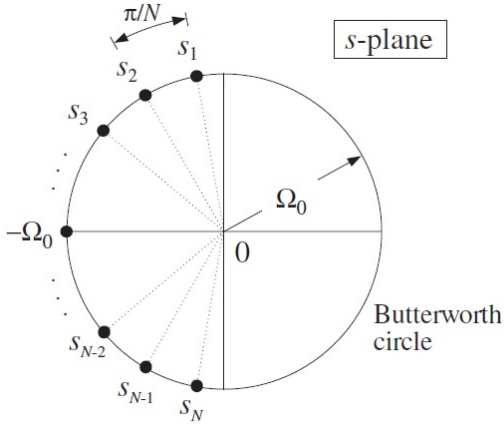
N must be an integer, we round N_{exact} up to the nearest integer: $N = \lceil N_{exact} \rceil$

$$\Omega_0 = \frac{\Omega_{pass}}{(10^{A_{pass}/10} - 1)^{1/2N}} = \frac{\Omega_{pass}}{\epsilon_{pass}^{1/N}}$$

Of course, Ω_0 can be calculated in terms of the stopband. Then the passband attenuation will be a bit smaller than necessary.

Spectral factorization The amplitude response $|H(\Omega)|^2$ now has to be transformed into a transfer function $H(\Omega)$. With $s = j\Omega$, the fact that $|H(\Omega)|^2 = H(\Omega)H(\Omega)^*$ and setting $H(s) = \frac{1}{D(s)}$, we have

$$D(s)D(-s)^* = 1 + (-1)^N \left(\frac{s}{\Omega_0}\right)^{2N}$$



To find an $H(s)$ that is causal and stable, we have to find all the roots in the left hand side of the s-plane of the following polynomial:

$$1 + (-1)^N \left(\frac{s}{\Omega_0}\right)^{2N} = 0 \quad \Rightarrow \quad s^{2N} = (-1)^{N-1} \Omega_0^{2N}$$

The $2N$ roots are equally spaced on a circle in the s-plane with the radius of Ω_0 .

$$s_i = \Omega_0 e^{j\theta_i}, \quad \theta_i = \frac{\pi}{2N}(N - 1 + 2i), \quad i = 1, 2, \dots, 2N$$

If N is even ($N = 2K$), there are exactly K conjugate complex root pairs. If N is odd ($N = 2K + 1$), there are k conjugate complex root pairs and one single root at $s = -\Omega_0$. The conjugate complex pairs can be written as second order section, allowing to write $H(s)$ as

$$H(s) = H_0(s)H_1(s)H_2(s) \dots H_K(s) \quad \text{where} \quad \begin{aligned} H_0(s) &= \begin{cases} 1, & \text{if } N = 2K \\ \frac{1}{1 + \frac{s}{\Omega_0}}, & \text{if } N = 2K + 1 \end{cases} \\ H_i(s) &= \frac{1}{1 - 2\frac{s}{\Omega_0} \cos \theta_i + \frac{s^2}{\Omega_0^2}}, \quad i = 1, 2, \dots, K \end{aligned}$$

A list of Butterworth polynomials can be found in appendix [C](#).

Design steps

1. Calculate the digital frequencies $\{\omega_{pass}, \omega_{stop}\}$ and the corresponding prewarped version $\{\Omega_{pass}, \Omega_{stop}\}$.
2. Calculate the order N and the 3-dB frequency Ω_0 of the equivalent analog Butterworth filter based on the transformed specifications $\{\Omega_{pass}, \Omega_{stop}, A_{pass}, A_{stop}\}$
3. Calculate the coefficients for all second-order sections and the transfer function.

Butterworth Design Recipe

1. Digital $\rightarrow$ analog frequencies

$$\omega = \frac{2\pi f}{f_s}, \quad \Omega = \tan \frac{\omega}{2}$$

2. Ripple factors

$$\epsilon = \sqrt{10^{A/10} - 1}$$

3. Filter order

$$N = \left\lceil \frac{\log(\epsilon_s/\epsilon_p)}{\log(\Omega_s/\Omega_p)} \right\rceil$$

4. Cutoff frequency

$$\Omega_0 = \frac{\Omega_p}{\epsilon_p^{1/N}}$$

5. Butterworth poles

$$p_k = \Omega_0 e^{j(\frac{\pi}{2} + \frac{2k+1}{2N}\pi)}, \quad k = 0, \dots, N-1$$

6. Form $H_a(s)$.

7. Apply the bilinear transformation

Replace every occurrence of s in the analog transfer function

$$H_a(s)$$

by the corresponding bilinear transformation

$$s = f(z).$$

Examples:

Lowpass

$$s = \frac{1 - z^{-1}}{1 + z^{-1}}$$

$$H(z) = H_a\left(\frac{1 - z^{-1}}{1 + z^{-1}}\right)$$

Bandpass

$$s = \frac{1 - 2cz^{-1} + z^{-2}}{1 - z^{-2}}$$

$$H(z) = H_a \left(\frac{1 - 2cz^{-1} + z^{-2}}{1 - z^{-2}} \right)$$

Finally, expand the numerator and denominator, multiply by the required factor (e.g. $(1 + z^{-1})^N$ or $(1 - z^{-2})^N$) to remove fractions, and collect equal powers of z^{-1} .

8. Implement as cascade of SOS.

$$\boxed{H_a(s) \implies s = f(z) \implies H(z) \implies \text{expand \& simplify}}$$

5.3.2. Digital Lowpass Filters S599

As before, the analog lowpass filter must be transformed into a digital filter using the bilinear transform

$$H_i(z) = \frac{1}{1 - 2\frac{s}{\Omega_0} \cos \theta_i + \frac{s^2}{\Omega_0^2}} \bigg|_{s=\frac{1-z^{-1}}{1+z^{-1}}} = \frac{G_i(1 + z^{-1})^2}{1 + a_{i1}z^{-1} + a_{i2}z^{-2}}$$

where the filter coefficients G_i, a_{i1}, a_{i2} are easily found to be

$$G_i = \frac{\Omega_0^2}{1 - 2\Omega_0 \cos \theta_i + \Omega_0^2} \quad a_{i1} = \frac{2(\Omega_0^2 - 1)}{1 - 2\Omega_0 \cos \theta_i + \Omega_0^2} \quad a_{i2} = \frac{1 + 2\Omega_0 \cos \theta_i + \Omega_0^2}{1 - 2\Omega_0 \cos \theta_i + \Omega_0^2}$$

If N is odd, there is of course a first-order section

$$H_0(z) = \frac{1}{1 + \frac{s}{\Omega_0}} \bigg|_{s=\frac{1-z^{-1}}{1+z^{-1}}} = \frac{G_0(1 + z^{-1})}{1 + a_{01}z^{-1}} \quad \text{with} \quad G_0 = \frac{\Omega_0}{\Omega_0 + 1}, \quad a_{01} = \frac{\Omega_0 - 1}{\Omega_0 + 1}$$

This results in the overall digital transfer function $H(z) = H_0(z)H_1(z)H_2(z) \dots H_K(z)$. The 3dB frequency f_0 in Hz is related to the Butterworth parameter Ω_0 over the bilinear transform:

$$f_0 = \frac{f_s}{\pi} \arctan(\Omega_0)$$

5.3.3. Digital Highpass Filters S603

The highpass version of the bilinear transform is used to map the given highpass specifications onto equivalent analog lowpass specifications.

For a highpass filter, the passband and stopband frequencies are calculated by

$$\Omega_{pass} = \cot \left(\frac{\omega_{pass}}{2} \right) = \cot \left(\frac{\pi f_{pass}}{f_s} \right)$$

$$\Omega_{stop} = \cot \left(\frac{\omega_{stop}}{2} \right) = \cot \left(\frac{\pi f_{stop}}{f_s} \right)$$

The highpass bilinear transform is then used to find $H(z)$

$$H_i(z) = \frac{G_i(1 - z^{-1})^2}{1 + a_{i1}z^{-1} + a_{i2}z^{-2}}$$

where

$$G_i = \frac{\Omega_0^2}{1 - 2\Omega_0 \cos \theta_i + \Omega_0^2} \quad a_{i1} = -\frac{2(\Omega_0^2 - 1)}{1 - 2\Omega_0 \cos \theta_i + \Omega_0^2} \quad a_{i2} = \frac{1 + 2\Omega_0 \cos \theta_i + \Omega_0^2}{1 - 2\Omega_0 \cos \theta_i + \Omega_0^2}$$

Again, if N is odd, then there is a first-order section

$$H_0(z) = \frac{G_0(1 - z^{-1})}{1 + a_{01}z^{-1}} \quad \text{where} \quad G_0 = \frac{\Omega_0}{\Omega_0 + 1}, \quad a_{01} = -\frac{\Omega_0 - 1}{\Omega_0 + 1}$$

The 3dB frequency f_0 of the highpass filter can then be calculated as

$$f_0 = \frac{f_s}{\pi} \arctan\left(\frac{1}{\Omega_0}\right)$$

Note the similarities and differences between the highpass and the lowpass case: The coefficients G_i and a_{i2} are the same, but a_{i1} has reverse sign. The numerator of the second order sections is now $(1 - z^{-1})^2$ instead of $(1 + z^{-1})^2$, resulting in a zero at $z = 1$ or $\omega = 0$.

5.3.4. Digital Bandpass Filters S606

The bandpass filter design works similarly. This design will result in the same attenuation in both stopbands. Hence one must focus on the stopband with the more stringent specification.

The bandpass specifications are mapped onto the entire passband $[-\Omega_{pass}, \Omega_{pass}]$ of the analog filter by

$$\Omega = \frac{c - \cos \omega}{\sin \omega} \quad \text{where} \quad c = \frac{\sin(\omega_{pa} + \omega_{pb})}{\sin \omega_{pa} + \sin \omega_{pb}}$$

Note that $c = \cos \omega_c$ where ω_c can be thought of as the center frequency of the bandpass filter. In general ω_c is *not* exactly the center of the passband!

This results in

$$\Omega_{pass} = \left| \frac{c - \cos \omega_{pb}}{\sin \omega_{pb}} \right| \quad \text{and} \quad \Omega_{stop} = \min(|\Omega_{sa}|, |\Omega_{sb}|) \quad \text{with} \quad \Omega_{sa} = \frac{c - \cos \omega_{sa}}{\sin \omega_{sa}}, \quad \Omega_{sb} = \frac{c - \cos \omega_{sb}}{\sin \omega_{sb}}$$

Again, $H_i(z)$ and, if N is odd also $H_0(z)$ are calculated as

$$H_i(z) = \frac{G_i(1 - z^{-2})^2}{1 + a_{i1}z^{-1} + a_{i2}z^{-2} + a_{i3}z^{-3} + a_{i4}z^{-4}} \quad \text{where} \quad G_i = \frac{\Omega_0^2}{1 - 2\Omega_0 \cos \theta_i + \Omega_0^2}$$

$$a_{i1} = \frac{4c(\Omega_0 \cos \theta_i - 1)}{1 - 2\Omega_0 \cos \theta_i + \Omega_0^2} \quad a_{i2} = \frac{2(2c^2 + 1 - \Omega_0^2)}{1 - 2\Omega_0 \cos \theta_i + \Omega_0^2}$$

$$a_{i3} = -\frac{4c(\Omega_0 \cos \theta_i + 1)}{1 - 2\Omega_0 \cos \theta_i + \Omega_0^2} \quad a_{i4} = \frac{1 + 2\Omega_0 \cos \theta_i + \Omega_0^2}{1 - 2\Omega_0 \cos \theta_i + \Omega_0^2}$$

and

$$H_0(z) = \frac{G_0(1 - z^{-2})}{1 + a_{01}z^{-1} + a_{02}z^{-2}} \quad \text{where}$$

$$G_0 = \frac{\Omega_0}{1 + \Omega_0}, \quad a_{01} = -\frac{2c}{1 + \Omega_0}, \quad a_{02} = \frac{1 - \Omega_0}{1 + \Omega_0}$$

Each filter section has zeros at $z = \pm 1$, i.e. $\omega = 0$ and $\omega = \pi$. The left and the right 3dB frequencies ω_{0a} and ω_{0b} can be calculated by

$$\tan\left(\frac{\omega_{0a}}{2}\right) = \frac{\sqrt{\Omega_0^2 + 1 - c^2} - \Omega_0}{1 + c} \quad \text{and} \quad \tan\left(\frac{\omega_{0b}}{2}\right) = \frac{\sqrt{\Omega_0^2 + 1 - c^2} + \Omega_0}{1 + c}$$

5.3.5. Digital Bandstop Filters S611

The design of digital bandstop filters is similar to that of digital bandpass filters.

$$c = \frac{\sin(\omega_{pa} + \omega_{pb})}{\sin \omega_{pa} + \sin \omega_{pb}}, \quad \Omega_{pass} = \left| \frac{\sin \omega_{pb}}{\cos \omega_{pb} - c} \right|$$

$$\Omega_{stop} = \min(|\Omega_{sa}|, |\Omega_{sb}|) \quad \text{where} \quad \Omega_{sa} = \frac{\sin \omega_{sa}}{\cos \omega_{sa} - c}, \quad \Omega_{sb} = \frac{\sin \omega_{sb}}{\cos \omega_{sb} - c}$$

The bilinear transform of the second order sections is then

$$H_i(z) = \frac{G_i(1 - 2cz^{-1} + z^{-2})^2}{1 + a_{i1}z^{-1} + a_{i2}z^{-2} + a_{i3}z^{-3} + a_{i4}z^{-4}} \quad \text{where} \quad G_i = \frac{\Omega_0^2}{1 - 2\Omega_0 \cos \theta_i + \Omega_0^2}$$

$$a_{i1} = \frac{4c\Omega_0(\cos \theta_i - \Omega_0)}{1 - 2\Omega_0 \cos \theta_i + \Omega_0^2} \quad a_{i2} = \frac{2(2c^2\Omega_0^2 + \Omega_0^2 - 1)}{1 - 2\Omega_0 \cos \theta_i + \Omega_0^2}$$

$$a_{i3} = \frac{4c\Omega_0(\cos \theta_i + \Omega_0)}{1 - 2\Omega_0 \cos \theta_i + \Omega_0^2} \quad a_{i4} = \frac{1 + 2\Omega_0 \cos \theta_i + \Omega_0^2}{1 - 2\Omega_0 \cos \theta_i + \Omega_0^2}$$

and

$$H_0(z) = \frac{G_0(1 - 2cz^{-1} + z^{-2})}{1 + a_{01}z^{-1} + a_{02}z^{-2}} \quad \text{where}$$

$$G_0 = \frac{\Omega_0}{1 + \Omega_0}, \quad a_{01} = -\frac{2c\Omega_0}{1 + \Omega_0}, \quad a_{02} = -\frac{1 - \Omega_0}{1 + \Omega_0}$$

Each second order section has zeros at $\omega = \pm\omega_c$, which makes sense for a bandstop filter. The 3dB frequencies are now

$$\tan\left(\frac{\omega_{0a}}{2}\right) = \frac{\sqrt{1 + \Omega_0^2(1 - c^2)} - 1}{\Omega_0(1 + c)} \quad \text{and} \quad \tan\left(\frac{\omega_{0b}}{2}\right) = \frac{\sqrt{1 + \Omega_0^2(1 - c^2)} + 1}{\Omega_0(1 + c)}$$

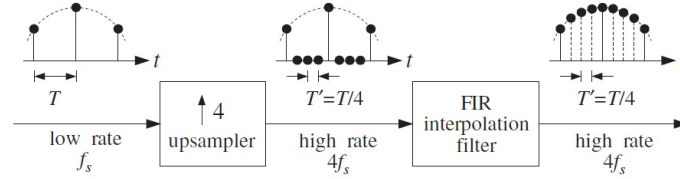
6. Interpolation, Decimation and OversamplingChapter 12

6.1. Interpolation and Oversampling S632-637

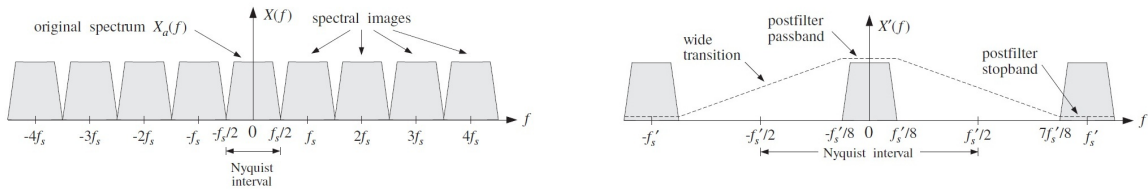
Oversampling is increasing the sample rate and requires some form of interpolation. New samples at a higher sampling rate are created. These are calculated using a FIR filter. The spectrum of the low-rate and the high-rate signal are identical, if the frequency axis is not normalized.

6.2. Interpolation Filter Design S638-657

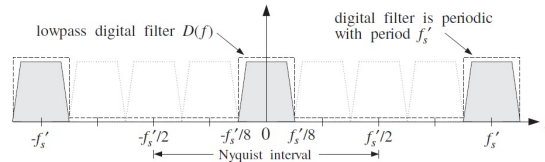
6.2.1. Direct form S638



There are now two discrete times, a fast one n' and a slow one called n . The effect on the spectrum of a signal and the advantage in DA-Conversion is shown in the following figure (left: original spectrum, right: spectrum at high rate).



The Job of the FIR filter is to move the zeros to the appropriate values, or to erase the spectral copies of the original spectrum in the new Nyquist band.



The ideal lowpass filter for an L -fold interpolation is operating at the fast sampling rate $f'_s = Lf_s$ and has its cutoff frequency at the low-sampling rate's Nyquist frequency $f_s/2$.

$$f_c = \frac{f_s}{2} = \frac{f'_s}{2L} \quad \omega'_c = \frac{2\pi f_c}{f'_s} = \frac{\pi}{L}$$

FIR approximation to the ideal interpolator (non causal):

By delaying this filter by LM samples, it gets causal:

Using a Hamming window:

Hamming window

$$d(k') = \frac{\sin(\pi k'/L)}{\pi k'/L}, \text{ with } -LM \leq k' \leq LM$$

$$h(n') = d(n' - LM) = \frac{\sin(\pi(n' - LM)/L)}{\pi(n' - LM)/L}$$

$$h(n') = w(n')d(n' - LM)$$

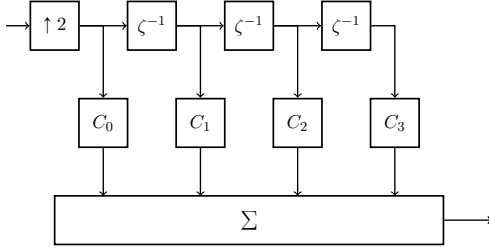
$$w(n') = 0.54 - 0.46 \cos\left(\frac{2\pi n'}{N-1}\right)$$

The output of the non causal FIR interpolator filter is obtained as the convolution of the upsampled signal with the filter

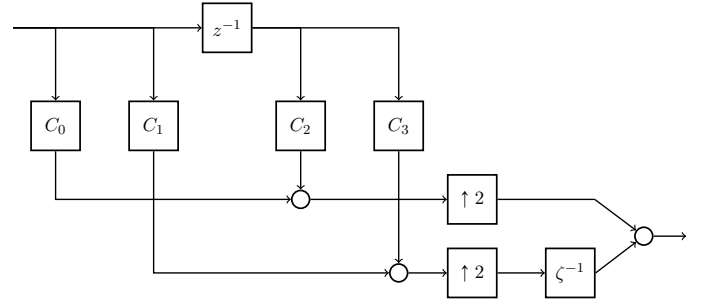
$$y_{up}(n') = \sum_{k'=-LM}^{LM} d(k')x_{up}(n' - k')$$

6.2.2. Polyphase Form S640

The direct form is



A more efficient way is using the polyphase form



Note that the direct form does all multiplications at the high rate, while the polyphase form allows doing the multiplications at the low rate, which makes the system more efficient.

Suppose the low-rate time instants are n , then the fast-rate time instants can be described as $n' = NL + i$ for $i = 0, 1, \dots, L - 1$. The output is therefore

$$y_{up}(nL + i) = \sum_{k=-M}^{M-1} \sum_{j=0}^{L-1} d(kL + j) x_{up}(nL + i - kL - j)$$

which allows us to split the output y_{up} into $y_i(n) = y_{up}(nL + i)$. We can also define the i th polyphase subfilter as

$$d_i(k) = d(kL + i), \quad -M \leq k \leq M - 1, \quad i = 0, 1, \dots, L$$

The i th outputs can now be calculated in terms of the slow-rate samples

$$y_i(n) = \sum_{k=-M}^{M-1} d_i(k) x(n - k), \quad i = 0, 1, \dots, L - 1$$

The output y_{up} is now only calculated in terms of the slow sampling rate, which decreases the computational rate in terms of multiplications per second by a factor of L , from $R = NLf_s$ (direct form) to $R = Nf_s$ (polyphase form).

6.2.3. Frequency domain characteristics S645

To discuss the interpolation in the frequency domain, we introduce the different unit delays z (low sampling rate = long delay) and ζ (high sampling rate = short delay)

$$z = \zeta^L \quad \Leftrightarrow \quad \zeta = z^{1/L}$$

The high-rate output can now be written in the z -Domain as

$$Y_{up}(\zeta) = \sum_{i=0}^{L-1} \zeta^{-i} Y_i(\zeta^L) = \sum_{i=0}^{L-1} z^{-i/L} Y_i(z)$$

which is what we already know from the time domain. The filter $D(\zeta)$ similarly is

$$D(\zeta) = \sum_{i=0}^{L-1} \zeta^{-i} D_i(\zeta^L) = \sum_{i=0}^{L-1} z^{-i/L} D_i(z)$$

Low-Rate Representation The original low-rate signal is represented in the z -domain by

$$Y(z) = H(z)X(z),$$

where the sampling frequency is

$$f_s.$$

After interpolation by the factor L , the sampling frequency becomes

$$f'_s = Lf_s,$$

and the corresponding transform variable changes from

$$z \longrightarrow \zeta.$$

The high-rate output is therefore represented by

$$Y_{up}(\zeta).$$

Overall High-Rate Transfer Function Replacing

$$z = \zeta^L$$

gives the high-rate interpolation filter

$$H(\zeta) = \sum_{i=0}^{L-1} \zeta^{-i} H_i(\zeta^L).$$

For $L = 4$,

$$H(\zeta) = H_0(\zeta^4) + \zeta^{-1} H_1(\zeta^4) + \zeta^{-2} H_2(\zeta^4) + \zeta^{-3} H_3(\zeta^4).$$

The 0th polyphase subfilter $d_0(k)$ should of course return the same value as the input $x(n)$, so the impulse response must be a dirac: $d_0(k) = d(kL) = \delta(k)$.

The ideal filter which removes all spectral copies would generally have the following characteristics

$$D(f) = \begin{cases} L, & \text{if } |f| \leq \frac{f_s}{2} \\ 0, & \text{if } \frac{f_s}{2} \leq |f| \leq \frac{f'_s}{2} \end{cases}$$

The ideal reconstructor $d(k)$, running at the higher rate f'_s would therefore be

$$d(k') = \frac{\sin(\pi k'/L)}{\pi k'/L}$$

This reconstructor is usually implemented using a Kaiser window design.

Low rate	High rate
f_s	$f'_s = Lf_s$
z	ζ
$Y(z)$	$Y_{up}(\zeta)$

6.2.4. Kaiser Design for Interpolation Filters

For an L -fold interpolator the Kaiser design equations are slightly modified, because the filter operates at the high sampling rate

$$f'_s = Lf_s.$$

The transition width is

$$\Delta f = f_{stop} - f_{pass},$$

or in normalized form

$$\Delta F = \frac{\Delta f}{f_s}.$$

The Kaiser parameters are

$$D = \frac{A - 7.95}{14.36}, \quad \alpha = 0.1102(A - 8.7).$$

The required filter length is

$$N - 1 \geq \frac{DL}{\Delta F}$$

where N is rounded up to the next odd integer of the form

$$N = 2LM + 1.$$

Hence

$$M = \frac{N - 1}{2L}.$$

The computational rate becomes

$$R_{\text{direct}} = NLf_s, \quad R_{\text{polyphase}} = Nf_s.$$

6.3. Linear and Hold Interpolators S657-661

The reconstructor described above is the ideal interpolator. Nevertheless there exist simpler interpolators, such as the linear and the hold interpolators. Not surprising, these are not as good as the ideal interpolator, but especially at high sample rates, the advantage of less calculations overweighs the downsides.

6.3.1. Hold Interpolator

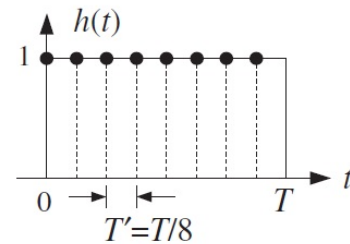
$$d(k') = \begin{cases} 1, & \text{if } 0 \leq k' \leq L - 1 \\ 0, & \text{otherwise} \end{cases}$$

$$y_{up}(nL + i) = x(n)$$

$$D(f) = \frac{\sin(\pi f / f_s)}{\sin(\pi f / Lf_s)} e^{-j\pi(L-1)f / (Lf_s)}$$

The polyphase subfilters $d_i(k)$ will then be

$$d_i(k) = \delta(k)$$

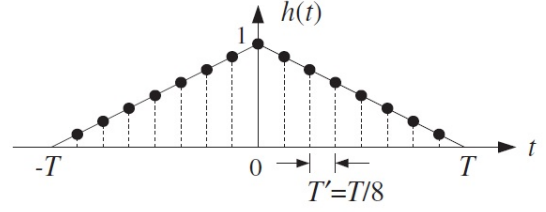


6.3.2. Linear Interpolator

$$d(k') = \begin{cases} 1 - \frac{|k'|}{L}, & \text{if } |k'| \leq L-1 \\ 0, & \text{otherwise} \end{cases}$$

$$y_{up}(nL + i) = (1 - \frac{i}{L})x(n) + \frac{i}{L}x(n+1)$$

$$D(f) = \frac{1}{L} \left| \frac{\sin(\pi f/f_s)}{\sin(\pi f/Lf_s)} \right|^2$$



The polyphase subfilters $d_i(k)$ will then be

$$d_i(k) = (1 - \frac{i}{L})\delta(k) + \frac{i}{L}\delta(k+1)$$

Replication Property The ideal interpolation filter satisfies

$$\frac{1}{L} \sum_{m=0}^{L-1} D(f - mf_s) = 1.$$

Thus, the passband gain of the ideal interpolation filter must be

$$L,$$

which compensates for the amplitude reduction introduced by the upsampler.

Interpretation For every frequency f , exactly one replicated spectrum lies inside the passband of the ideal interpolation filter.

Therefore,

$$\sum_{m=0}^{L-1} D(f - mf_s) = L,$$

and after scaling by $1/L$,

$$\frac{1}{L} \sum_{m=0}^{L-1} D(f - mf_s) = 1.$$

Hence the overall gain remains unity although the ideal interpolation filter has a passband gain of L .

6.4. Design Examples S661-686

6.4.1. 4-fold interpolators S661

$$L = 4$$

with polyphase filter length $2M = 4$ or $M = 2$ which leads to a filter length $N = 2LM + 1 = 17$. The ideal impulse response is:

$$d(k') = \frac{\sin(\pi k'/4)}{\pi k'/4}, \quad -8 \leq k' \leq 8$$

or, numerically,

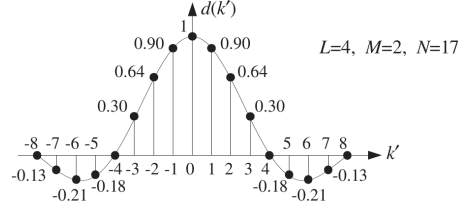
$$\mathbf{h} = \mathbf{d} = [0, -0.13, -0.21, -0.18, 0, 0.30, 0.64, 0.90, 1, 0.90, 0.64, 0.30, 0, -0.18, -0.21, -0.13, 0]$$

where $\mathbf{h}$ is the causal version, and $\mathbf{d}$ is the symmetric one with time origin at the middle of the vector. The resulting polyphase subfilters

$$d_i(k) = d(4k + i), \quad -2 \leq k \leq 1$$

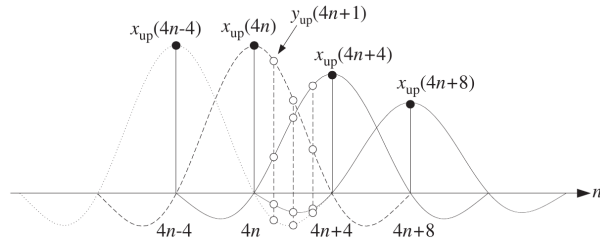
can now be calculated (high-rate filter subsampled by a factor of 4 and an increasing offset)

$$\begin{aligned} \mathbf{h}_0 &= \mathbf{d}_0 = [0, 0, 1, 0] \\ \mathbf{h}_1 &= \mathbf{d}_1 = [-0.13, 0.30, 0.90, -0.18] \\ \mathbf{h}_2 &= \mathbf{d}_2 = [-0.21, 0.64, 0.64, -0.21] \\ \mathbf{h}_3 &= \mathbf{d}_3 = [-0.18, 0.90, 0.30, -0.13] \end{aligned}$$



The interpolated samples $y_{\text{up}}(4n + i)$ between $x(n) = x_{\text{up}}(4n)$ and $x(n + 1) = x_{\text{up}}(4[n + 1])$ are now:

$$\begin{bmatrix} y_{\text{up}}(4n) \\ y_{\text{up}}(4n + 1) \\ y_{\text{up}}(4n + 2) \\ y_{\text{up}}(4n + 3) \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ -0.13 & 0.30 & 0.90 & -0.18 \\ -0.21 & 0.64 & 0.64 & -0.21 \\ -0.18 & 0.90 & 0.30 & -0.13 \end{bmatrix} \begin{bmatrix} x_{\text{up}}(4n + 8) \\ x_{\text{up}}(4n + 4) \\ x_{\text{up}}(4n) \\ x_{\text{up}}(4n - 4) \end{bmatrix}$$



The 17 sample long FIR filter is very short ($M = 2$). This and the rectangular window have the effect that the interpolated signal is of low quality. Rectangular windows should not be used, Hamming is much better and results in a reasonable interpolation even with $M = 2$

The Hamming windowed version is obtained by multiplying the high-rate 17 sample long filter with the appropriate Hamming window resulting in

$$\begin{bmatrix} y_{\text{up}}(4n) \\ y_{\text{up}}(4n + 1) \\ y_{\text{up}}(4n + 2) \\ y_{\text{up}}(4n + 3) \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ -0.02 & 0.22 & 0.87 & -0.07 \\ -0.05 & 0.55 & 0.55 & -0.05 \\ -0.07 & 0.87 & 0.22 & -0.02 \end{bmatrix} \begin{bmatrix} x_{\text{up}}(4n + 8) \\ x_{\text{up}}(4n + 4) \\ x_{\text{up}}(4n) \\ x_{\text{up}}(4n - 4) \end{bmatrix}$$

Important Property For ideal interpolation

$$d_0(k) = \delta(k)$$

therefore

$$y_{\text{up}}(Ln) = x(n).$$

Only the inserted samples are interpolated.

The resulting magnitude response shows, that the rectangular filter is steeper, at the high cost of 8.9% overshoot, which the Hamming window version doesn't have.

For a block diagram realization of the polyphase form, recall these two relationships:

$$Y_{\text{up}}(\xi) = \sum_{i=0}^{L-1} \xi^{-i} Y_i(\xi^L) = \sum_{i=0}^{L-1} z^{-i/L} Y_i(z)$$

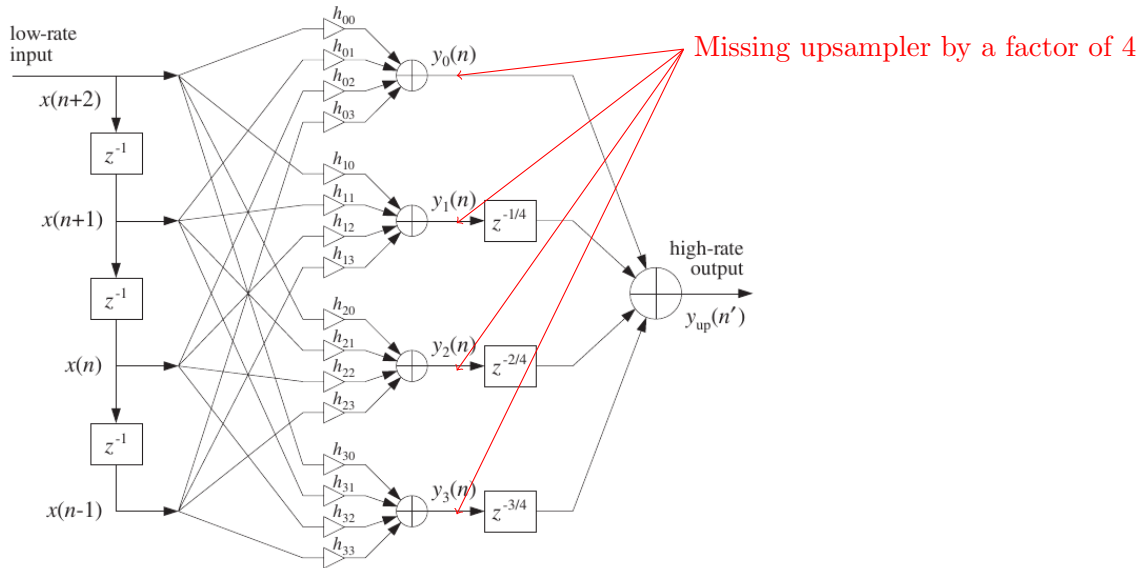
$$D(\xi) = \sum_{i=0}^{L-1} \xi^{-i} D_i(\xi^L) = \sum_{i=0}^{L-1} z^{-i/L} D_i(z)$$

Therefore the transfer function of the high-rate interpolator filter can be written as the sum of the polyphase subfilters appropriately delayed:

$$H(\xi) = H_0(\xi^4) + \xi^{-1} H_1(\xi^4) + \xi^{-2} H_2(\xi^4) + \xi^{-3} H_3(\xi^4)$$

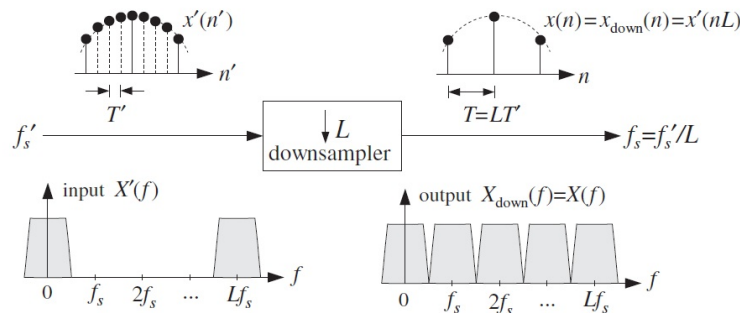
$$= H_0(z) + z^{-1/4} H_1(z) + z^{-2/4} H_2(z) + z^{-3/4} H_3(z)$$

The resulting block diagram clearly shows how the four polyphase filters create the four output values. Three of these values are interpolated while value $y_{\text{up}}(n') = y_{\text{up}}(4n) = y_0(n)$ is identical to the upsampled input value $x_{\text{up}}(n') = x_{\text{up}}(4n) = x(n)$, since $h_0 = [0, 0, 1, 0]$. Clearly, the four output samples can be calculated in parallel, allowing for a very fast implementation.



6.5. Decimation and Oversampling S686-691

Decimation is the inverse of oversampling, thus the sampling rate is reduced from a high rate f'_s to a low rate $f_s = f'_s/L$.



Downsampling is simply the process of discarding $L - 1$ samples:

$$x_{\text{down}}(n) = x'(n')|_{n'=nL} = x'(nL)$$

Frequency-Domain Relation Downsampling causes a periodic replication of the high-rate spectrum:

$$X(f) = \frac{1}{L} \sum_{m=0}^{L-1} X'(f - mf_s).$$

Using normalized digital frequency

$$\omega = \frac{2\pi f}{f_s},$$

this becomes

$$X(\omega) = \frac{1}{L} \sum_{m=0}^{L-1} X' \left(\omega' - \frac{2\pi m}{L} \right).$$

Proof Idea (Problem 12.12) Starting from the two Poisson summation formulas

$$X(f) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_a(f - kf_s), \quad X'(f) = \frac{1}{T'} \sum_{k'=-\infty}^{\infty} X_a(f - k'f'_s),$$

use the substitution

$$k = Lk' + m, \quad m = 0, \dots, L-1,$$

together with

$$f'_s = Lf_s, \quad T = LT',$$

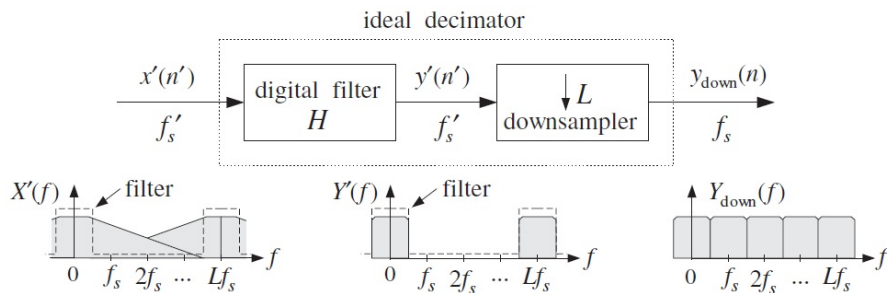
to obtain

$$X(f) = \frac{1}{L} \sum_{m=0}^{L-1} X'(f - mf_s).$$

The factor $1/L$ compensates for the sum of the L replicated spectra.

Caution: if the high rate spectrum occupies more than the low rate Nyquist rate there *will* be aliasing.

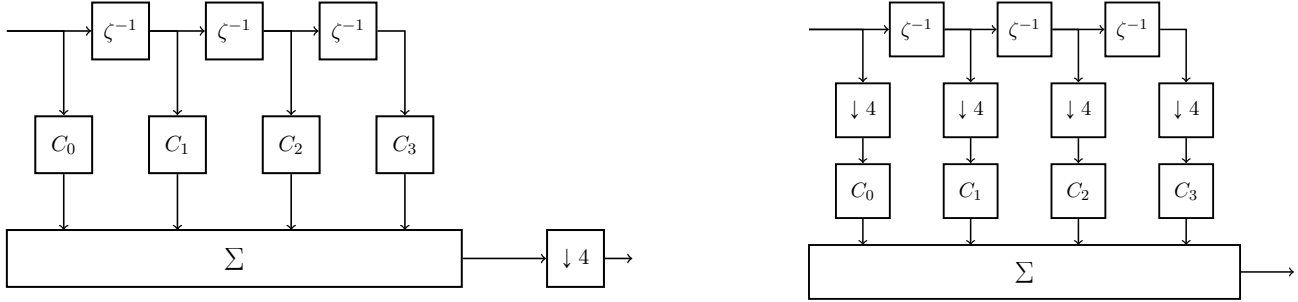
The ideal *decimator* therefore contains a decimation filter, which operates at the high frequencies and cuts out the low rate Nyquist band.



The decimation filter is similar to the interpolation filter, except that its DC gain is unity instead of L . A length- N FIR decimation filter can therefore be obtained by

$$h(n') = w(n')d(n' - LM), \quad \text{where} \quad d(k') = \frac{\sin(\pi k'/L)}{\pi k'}$$

Since only every L th output sample is needed, the filter output does not need to be calculated for every fast rate sample (like on the left image), reducing the computational cost by a factor L to $R = \frac{1}{L}Nf'_s = Nf_s$ (left image).



Multistage designs are very popular for decimation filters. The early filters will then be fast but simple, while later filters are slow but very precise. For fast signals, often a simple FIR averaging filter is used. The last, slowest filter can then be designed to fix the overall frequency response.

Butterworth Attenuation For an analog Butterworth filter

$$A(F) = 10 \log_{10} \left(1 + \left(\frac{F}{F_0} \right)^{2N} \right),$$

where

$$F = \frac{f}{f_s}, \quad F_0 = \frac{f_0}{f_s}.$$

For oversampling prefilters

$$F_{pass} = 0.5, \quad F_{stop} = L - 0.5.$$

Oversampling Ratio For a third-order Butterworth analog antialiasing prefilter,

$$L \approx 0.94 \cdot 10^{A_{stop}/60} + 0.5$$

gives the minimum oversampling ratio required to achieve a stopband attenuation of A_{stop} .

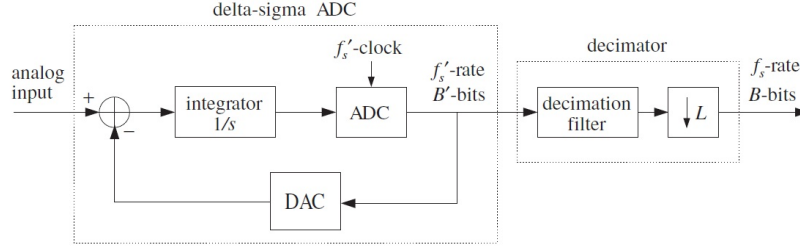
Butterworth Prefilter Order The required order of the analog Butterworth antialiasing filter is

$$N = \frac{\ln \left(\frac{10^{A_{stop}/10} - 1}{10^{A_{pass}/10} - 1} \right)}{2 \ln(2L - 1)}$$

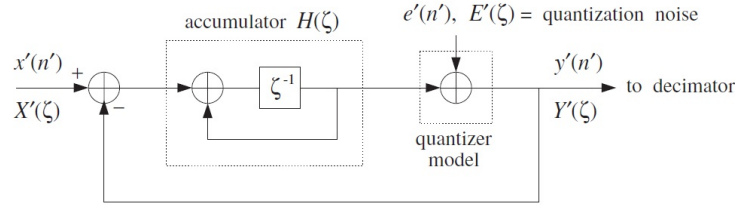
The filter order decreases as the oversampling factor L increases.

6.6. Noise Shaping Quantizer S698-705

First-Order Delta-Sigma A/D Converters For first-order delta-sigma A/D converters, only one output bit is used ($B' = 1$). By oversampling the input signal, the quantization noise is shifted towards higher frequencies, where it is removed by the digital decimation filter.



To obtain a discrete-time model, the quantizer is replaced by an equivalent additive quantization noise source.



General Signal and Noise Transfer Functions For a loop filter $H(\zeta)$, the input and noise transfer functions are

$$H_X(\zeta) = \frac{H(\zeta)}{1 + H(\zeta)}$$

$$H_{NS}(\zeta) = \frac{1}{1 + H(\zeta)}$$

These relations are valid for arbitrary noise-shaping loops.

First-Order Noise Shaping For the first-order accumulator

$$H(\zeta) = \frac{\zeta^{-1}}{1 - \zeta^{-1}}$$

the transfer functions become

$$H_X(\zeta) = \zeta^{-1}$$

$$H_{NS}(\zeta) = 1 - \zeta^{-1}.$$

Higher-Order Noise Shaping For a noise shaper of order p

$$H_{NS}(\zeta) = (1 - \zeta^{-1})^p.$$

For the second-order case,

$$H_{NS}(\zeta) = (1 - \zeta^{-1})^2.$$

Quantization Noise Power The mean-square quantization noise is

$$\sigma_e^2 = \sigma_{e'}^2 \frac{1}{f_s'} \int_{-f_s'/2}^{f_s'/2} |H_{NS}(f)|^2 df,$$

where

$$f_s' = Lf_s.$$

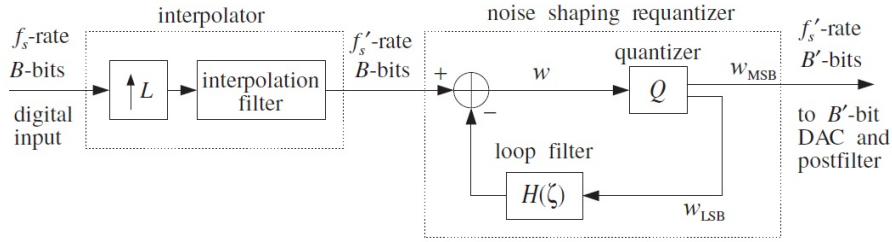
The magnitude response of the noise shaping filter is

$$|H_{NS}(f)|^2 = \left(\frac{\pi^{2p}}{2p+1} \right).$$

This leads to the well-known resolution improvement

$$\Delta B = (p + 0.5) \log_2 L - \frac{1}{2} \log_2 \left(\frac{\pi^{2p}}{2p+1} \right).$$

Oversampled Noise-Shaping Requantizers for D/A Conversion The same principle can also be applied to D/A conversion.



Typical loop filters are

$$\text{First-order: } H(\zeta) = \zeta^{-1}, \quad H_{NS}(\zeta) = 1 - \zeta^{-1},$$

$$\text{Second-order: } H(\zeta) = 2\zeta^{-1} - \zeta^{-2}, \quad H_{NS}(\zeta) = (1 - \zeta^{-1})^2.$$

The achievable resolution improvement is again

$$\Delta B = (p + 0.5) \log_2 L - \frac{1}{2} \log_2 \left(\frac{\pi^{2p}}{2p+1} \right).$$

A. DFT Matrices

$$W_2 = e^{-2\pi j/2} = -1$$

$$A_2 = \begin{bmatrix} 1 & 1 \\ 1 & W_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$W_4 = e^{-2\pi j/4} = \cos(\pi/2) - j \sin(\pi/2) = -j$$

$$A_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & W_4 & W_4^2 & W_4^3 \\ 1 & W_4^2 & W_4^4 & W_4^6 \\ 1 & W_4^3 & W_4^6 & W_4^9 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

$$W_8 = e^{-2\pi j/8} = \cos(\pi/4) - j \sin(\pi/4) = \frac{1-j}{\sqrt{2}}$$

$$A_8 = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & W_8 & W_8^2 & \dots & W_8^7 \\ 1 & W_8^2 & W_8^4 & \dots & W_8^{14} \\ 1 & W_8^3 & W_8^6 & \dots & W_8^{21} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & W_8^7 & W_8^{14} & \vdots & W_8^{49} \end{bmatrix}$$

$$W_8^0 = 1$$

$$W_8^1 = \frac{1-j}{\sqrt{2}}$$

$$W_8^2 = -j$$

$$W_8^3 = -\frac{1+j}{\sqrt{2}}$$

$$W_8^4 = -1$$

$$W_8^5 = -\frac{1-j}{\sqrt{2}}$$

$$W_8^6 = j$$

$$W_8^7 = \frac{1+j}{\sqrt{2}}$$

B. Common Pole / Zero Forms

The poles are obtained from the denominator

$$D(z) = 0,$$

and the zeros from the numerator

$$N(z) = 0.$$

For second-order polynomials the following standard forms occur frequently.

Polynomial	Pole / Zero Location	Comments
$1 - az^{-1}$	$z = a$	Real pole/zero.
$1 - 2\cos(\omega_0)z^{-1} + z^{-2}$	$z = e^{\pm j\omega_0}$	Complex-conjugate zeros on the unit circle ($ z = 1$).
$1 - 2r\cos(\omega_0)z^{-1} + r^2z^{-2}$	$z = re^{\pm j\omega_0}$	Complex-conjugate poles/zeros with radius r .
$1 - 2b\cos(\omega_0)z^{-1} + (2b - 1)z^{-2}$	Solve using the quadratic formula	Appears in notch and peak filters.

Useful Identities Euler's identity

$$e^{j\omega} = \cos(\omega) + j\sin(\omega)$$

implies

$$e^{-j\omega} = \cos(\omega) - j\sin(\omega).$$

Therefore

$$(z - e^{j\omega_0})(z - e^{-j\omega_0}) = z^2 - 2\cos(\omega_0)z + 1$$

or equivalently

$$1 - 2\cos(\omega_0)z^{-1} + z^{-2} = (1 - e^{j\omega_0}z^{-1})(1 - e^{-j\omega_0}z^{-1})$$

Similarly,

$$1 - 2r\cos(\omega_0)z^{-1} + r^2z^{-2} = (1 - re^{j\omega_0}z^{-1})(1 - re^{-j\omega_0}z^{-1})$$

Hence

$$|z| = \begin{cases} 1, & \text{unit circle} \\ r, & \text{radius } r \end{cases}$$

and

$$\angle z = \pm\omega_0.$$

Recognition Rules

$$1 - 2\cos(\omega_0)z^{-1} + z^{-2}$$

↓

- radius = 1
- angle = $\pm\omega_0$

—

$$1 - 2r\cos(\omega_0)z^{-1} + r^2z^{-2}$$

↓

- radius = r
- angle = $\pm\omega_0$

$$1 - az^{-1}$$

↓

- real pole/zero at $z = a$

C. Butterworth polynomials

N	K	$\theta_1, \theta_2, \dots, \theta_K$	$D(s)$
1	0		$1 + s$
2	1	$\frac{3\pi}{4}$	$(1 + 1.4142s + s^2)$
3	1	$\frac{4\pi}{6}$	$(1 + s)(1 + s + s^2)$
4	2	$\frac{5\pi}{8}, \frac{7\pi}{8}$	$(1 + 0.7654s + s^2)(1 + 1.8478s + s^2)$
5	2	$\frac{6\pi}{10}, \frac{8\pi}{10}$	$(1 + s)(1 + 0.6180s + s^2)(1 + 1.6180s + s^2)$
6	3	$\frac{7\pi}{12}, \frac{9\pi}{12}, \frac{11\pi}{12}$	$(1 + 0.5176s + s^2)(1 + 1.4142s + s^2)(1 + 1.9319s + s^2)$
7	3	$\frac{8\pi}{14}, \frac{10\pi}{14}, \frac{12\pi}{14}$	$(1 + s)(1 + 0.4450s + s^2)(1 + 1.2470s + s^2)(1 + 1.8019s + s^2)$

D. Kaiser Window

$$\alpha = \begin{cases} 0, & A < 21 \\ 0.5842(A - 21)^{0.4} + 0.07886(A - 21), & 21 \leq A \leq 50 \\ 0.1102(A - 8.7), & A > 50 \end{cases}$$

$$D = \begin{cases} 0.9222, & A < 21 \\ \frac{A - 7.95}{14.36}, & A \geq 21 \end{cases}$$

E. Useful DSP Relations

$$\omega = \frac{2\pi f}{f_s}$$

$$\Delta f = \frac{f_s}{N}$$

$$f'_s = Lf_s$$

$$f'_s = \frac{f_s}{M}$$

$$z = \zeta^L$$

F. Useful Transform Pairs

Signal	Transform
$\delta(n)$	1
$\delta(n - k)$	z^{-k}
$u(n)$	$\frac{1}{1 - z^{-1}}$
$a^n u(n)$	$\frac{1}{1 - az^{-1}}$
$na^n u(n)$	$\frac{az^{-1}}{(1 - az^{-1})^2}$

G. Idiotenseite

G.1. Funktionswerte für Winkelargumente

deg	rad	sin	cos	tan	deg	rad	sin	cos	deg	rad	sin	cos	deg	rad	sin	cos
0 °	0	0	1	0	90 °	$\frac{\pi}{2}$	1	0	180 °	π	0	-1	270 °	$\frac{3\pi}{2}$	-1	0
30 °	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$	120 °	$\frac{2\pi}{3}$	$\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	210 °	$\frac{7\pi}{6}$	$-\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	300 °	$\frac{5\pi}{3}$	$-\frac{\sqrt{3}}{2}$	$\frac{1}{2}$
45 °	$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1	135 °	$\frac{3\pi}{4}$	$\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$	225 °	$\frac{5\pi}{4}$	$-\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$	315 °	$\frac{7\pi}{4}$	$-\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$
60 °	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$	150 °	$\frac{5\pi}{6}$	$\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	240 °	$\frac{4\pi}{3}$	$-\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	330 °	$\frac{11\pi}{6}$	$-\frac{1}{2}$	$\frac{\sqrt{3}}{2}$

G.2. Quadrantenbeziehungen

$$\begin{aligned}
 \sin(-a) &= -\sin(a) & \cos(-a) &= \cos(a) \\
 \sin(\pi - a) &= \sin(a) & \cos(\pi - a) &= -\cos(a) \\
 \sin(\pi + a) &= -\sin(a) & \cos(\pi + a) &= -\cos(a) \\
 \sin\left(\frac{\pi}{2} - a\right) &= \sin\left(\frac{\pi}{2} + a\right) = \cos(a) & \cos\left(\frac{\pi}{2} - a\right) &= -\cos\left(\frac{\pi}{2} + a\right) = \sin(a)
 \end{aligned}$$

G.3. Periodizität

$$\cos(a + k \cdot 2\pi) = \cos(a) \quad \sin(a + k \cdot 2\pi) = \sin(a) \quad (k \in \mathbb{Z})$$

G.4. Additionstheoreme

$$\begin{aligned}
 \sin(a \pm b) &= \sin(a) \cdot \cos(b) \pm \cos(a) \cdot \sin(b) \\
 \cos(a \pm b) &= \cos(a) \cdot \cos(b) \mp \sin(a) \cdot \sin(b) \\
 \tan(a \pm b) &= \frac{\tan(a) \pm \tan(b)}{1 \mp \tan(a) \cdot \tan(b)}
 \end{aligned}$$

G.5. Doppel- und Halbwinkel

$$\begin{aligned}
 \sin(2a) &= 2 \sin(a) \cos(a) \\
 \cos(2a) &= \cos^2(a) - \sin^2(a) = 2 \cos^2(a) - 1 = 1 - 2 \sin^2(a) \\
 \cos^2\left(\frac{a}{2}\right) &= \frac{1 + \cos(a)}{2} & \sin^2\left(\frac{a}{2}\right) &= \frac{1 - \cos(a)}{2}
 \end{aligned}$$

G.6. Produkte

$$\begin{aligned}
 \sin(a) \sin(b) &= \frac{1}{2}(\cos(a - b) - \cos(a + b)) \\
 \cos(a) \cos(b) &= \frac{1}{2}(\cos(a - b) + \cos(a + b)) \\
 \sin(a) \cos(b) &= \frac{1}{2}(\sin(a - b) + \sin(a + b))
 \end{aligned}$$

G.7. Summe und Differenz

$$\begin{aligned}
 \sin(a) + \sin(b) &= 2 \cdot \sin\left(\frac{a+b}{2}\right) \cdot \cos\left(\frac{a-b}{2}\right) \\
 \sin(a) - \sin(b) &= 2 \cdot \sin\left(\frac{a-b}{2}\right) \cdot \cos\left(\frac{a+b}{2}\right) \\
 \cos(a) + \cos(b) &= 2 \cdot \cos\left(\frac{a+b}{2}\right) \cdot \cos\left(\frac{a-b}{2}\right) \\
 \cos(a) - \cos(b) &= -2 \cdot \sin\left(\frac{a+b}{2}\right) \cdot \sin\left(\frac{a-b}{2}\right) \\
 \tan(a) \pm \tan(b) &= \frac{\sin(a \pm b)}{\cos(a) \cos(b)}
 \end{aligned}$$

G.8. Euler-Formeln

Euler Identity

$$e^{j\varphi} = \cos \varphi + j \sin \varphi$$

Complex Numbers

$$e^{-j\varphi} = \cos \varphi - j \sin \varphi$$

$$(a + jb)^* = a - jb$$

Trigonometrische Darstellung

$$\sin \varphi = \frac{e^{j\varphi} - e^{-j\varphi}}{2j}, \quad \cos \varphi = \frac{e^{j\varphi} + e^{-j\varphi}}{2}$$

$$|a + jb| = \sqrt{a^2 + b^2}$$

Exponential Rules

$$e^{x+jy} = e^x e^{jy} = e^x (\cos y + j \sin y)$$

$$\angle(a + jb) = \text{atan2}(b, a)$$

$$e^{j\pi} = e^{-j\pi} = -1, \quad e^{j\omega} e^{-j\omega} = 1$$

$$|e^{j\omega}| = 1$$

G.9. Komplexe Zahlen

$$z = a + jb$$

$$z^* = a - jb$$

$$|z| = \sqrt{a^2 + b^2}$$

$$\angle z = \text{atan2}(b, a)$$

$$z = |z| e^{j\angle z}$$

G.10. DFT-Hilfen

$$W_N = e^{-j\frac{2\pi}{N}}$$

$$W_N^N = 1$$

$$W_N^{k+N} = W_N^k$$

$$\sum_{k=0}^{N-1} W_N^{mk} = \begin{cases} N, & m = 0 \\ 0, & m \neq 0 \end{cases}$$

G.11. Reihenentwicklungen

$$\text{Geometrische Reihe} \quad \sum_{n=0}^{\infty} x^n = \frac{1}{1-x} \quad |x| < 1$$

$$\sum_{k=0}^{\infty} k x^k = x \sum_{k=1}^{\infty} k x^{k-1} = \frac{x}{(1-x)^2} \quad x \neq 1$$

$$\sum_{k=0}^n a_0 \cdot q^k = a_0 \frac{1-q^{n+1}}{1-q} \quad q \neq 1$$

$$\text{Binominalreihe} \quad \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n = (1+x)^\alpha \quad x \in (-1, 1)$$

$$\text{E-Funktion} \quad \sum_{k=0}^{\infty} \frac{x^k}{k!} = e^x$$